The 15:50 Candle: A Pre-Registered Study of an End-of-Day Signal in 0DTE SPX Options*

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Abstract

I test whether a one-minute candle near the close predicts the move of the S&P 500 index (SPX) into the close, and whether the prediction is worth anything after the cost of trading same-day-expiry (0DTE) index options. Every confirmatory test was pre-registered. A scan of 41 candle features on 2013–2020 selected a 15:50 ET candle whose body spans at least 60% of its range. On the 2021–2026 holdout, traded in its own direction, it was right on 57.5% of 644 trades. At real OPRA quotes, an SPXW option 10 points in the money, bought at the ask and held to settlement, earned $67 a trade (Newey–West $t = 2.40$), about two-thirds of what a pricing model implied. The profit depends on the bid-ask spread, and the same trades lost money before 2021, when spreads were about five times wider. Early exits (including 26 pre-registered stop and take-profit rules at real bids), limit orders, momentum and dealer-gamma filters and smaller contracts did not improve the trade. The result also depends on where the candle is read. Read from the SPX index candle instead of the CFD candle of the backtests, the rule made $39 a trade ($t = 1.53$) and missed its criterion. Read from the SPY candle it made $72 ($t = 2.35$) and met a retrospective one. A final search over 7,128 settings of candle, cut, direction and strike again chose the 15:50 candle, with a luck diagnostic of $p = 0.051$. In a pre-registered test using the exchanges' imbalance messages, the candle pointed the same way as the closing-auction imbalance published at 15:50 on 73% of signal days but was not more accurate on those days (56% against 62%). I read the evidence as a small edge that is sensitive to execution and not yet established.

JEL: G13, G14
Keywords: 0DTE options; S&P 500 index options; closing auction; intraday momentum; bid-ask spread; pre-registration.

1 Introduction

Options on the S&P 500 index that expire the same day (0DTE) are now listed every weekday, and much of their volume concentrates near the close. The closing auction also becomes visible in the last ten minutes of the session. From 15:50 ET the New York Stock Exchange publishes the closing-auction imbalance of the stocks it lists, and Nasdaq disseminates an early imbalance indicator for its own stocks. An index option bought at 15:51 ET settles in

*This version: 12 November 2026. Every confirmatory test in this paper was pre-registered: each plan was hashed and time-stamped before the data it governs was read, and the tests that failed are reported with those that passed. This is the full working-paper version of the study and reports every study in full; it serves as the Internet Appendix to the shorter journal version of this paper. A statement on data availability is at the end of the paper. I thank Tiger Chen, whose observation started this study. This paper is research, not investment advice.

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cash on the official close nine minutes later, so its outcome depends on little more than the direction and size of the last move of the day. The close is therefore a reasonable place to look for a short-horizon directional signal, and a 0DTE option is the simplest way to trade one.

I came to this question through a friend, Tiger Chen, who had noticed by chance, on one-minute charts of the index, that the S&P 500 often makes a sharp move in one direction about ten minutes before the close, and who showed me 27 screenshots of it. This paper began as an attempt to check whether that observation is real and whether trading it with same-day options makes money.

A backtest of such a trade can mislead in several ways. If enough candle rules are scored on enough history, one of them will look good by chance, and its edge will shrink out of sample (Harvey et al., 2016). A backtest that enters at the close of the signal candle uses information that was not yet available. And a rule that is right 54% of the time can still lose money once the bid-ask spread is paid. I therefore fixed what would be tested before reading the data, and the confirmation prices the trade at quotes that were available at the time.

Every confirmatory test in the paper was pre-registered (Nosek et al., 2018). Before reading the data for a study I wrote down what would be computed, on which data, and what would count as success, and the plan was hashed and time-stamped. The tests that failed are reported with those that passed. The studies were done in sequence, each building on the last. A scan of 41 candle features finds a signal on 2013–2020 and tests it on 2021–2026 (Section 4). A calibrated pricing model then asks which strike best expresses the signal and whether a stop-loss or take-profit beats holding to settlement (Section 5). Real OPRA quotes replace the model to tune and confirm the trade at the prices actually offered (Section 6). Trade bars describe how the order might have been filled, and a pre-registered study tests real-time exit rules at real bids (Section 7). Three further studies re-tune the rule on recent data (Section 8), test published intraday momentum and a dealer-gamma channel (Section 9), and trade the signal on smaller contracts (Section 10). An independent re-validation then checks the whole chain (Section 11). Because the backtests read the signal from a contract for difference (CFD) that tracks the index, two further pre-registered studies read the signal from the SPX index candle itself (Section 12) and from the SPY candle (Section 13), and another searches 7,128 nearby settings, including other closing candles, for a better rule (Section 14). A last study tests an explanation of the signal with the closing-auction imbalances that the exchanges publish at 15:50 (Section 15).

The scan produced one usable signal, the 15:50 one-minute candle with a *strong body*, traded in its own direction. On 2021–2026 it was right on 57.5% of 644 trades. It is a stricter version of an effect I had already seen in earlier exploratory work, and it has weakened since 2023.

At real option prices the edge is small. Deep in-the-money options did best, in the model and at real prices. An option 10 points in the money, bought at the quoted ask and held to settlement, earned +\$67.46 per contract on 2021–2026 ($t = 2.40$), about two-thirds of the model's estimate. Whether the trade makes money depends mostly on the bid-ask spread. The same trades lost money at the ask before 2021, and a limit order at the mid does not recover the spread, because it misses the best days. No early exit improved on holding to settlement, including 26 stop and take-profit rules tested at real bids. Re-tuned thresholds,

momentum filters and cheaper contracts did not improve on the original rule either.

The result also depends on where the candle is read. An independent re-validation reproduced the results above, placed the start of the effect near a 60% body, and found that a larger body beyond that threshold does not mean a larger gain. It also showed that the CFD candle from which the signals were computed reads weaker than the SPX index candle. Read from the index candle, under a plan locked before any index candle was joined with a price, the rule made +$39.44 a trade ($t = 1.53$) and did not meet its criterion. The gain was confined to days on which both candles showed a strong body, so the confirmation above is a statement about the CFD candle. Read from the SPY candle, which can be seen live, the rule made +$71.74 a trade ($t = 2.35$) and met its criterion. SPY was the third source evaluated on the same outcomes, however, so this result is retrospective and is not an independent confirmation, and the gain was again confined to sessions on which all three candles showed a strong body.

A later pre-registered search over 7,128 settings of candle source, minute, cut, direction and strike chose the 15:50 candle again. Its choice was not clearly beyond luck ($p = 0.051$), and because 15:50 settings had already been examined on the later years, it could not be tested afresh. Finally, the candle pointed the same way as the closing-auction imbalance published at 15:50 on 73% of signal days, but in a pre-registered test it was not more accurate on those days (56% against 62%). The explanation that traders react to the published imbalance was therefore not confirmed, although the test could detect only a large difference.

Several limits apply throughout, and none of this is a recommendation to trade. The holdout was not blind, because a related effect was already known. The strike and exit results rest on a model calibrated on seven days. The real-price confirmation reuses index outcomes that I had already seen, and the execution results are descriptive proxies. The confirmed rule reads its candle from a CFD and not from the index, and the SPY candle's pass is retrospective. Section 17.3 discusses these limitations.

Related literature. The closest work is on market intraday momentum. Gao et al. (2018) find that the first half-hour return predicts the last half hour of the S&P 500, and Baltussen et al. (2021) find the same pattern across futures markets and link it to the hedging of option dealers who are short gamma. The 15:50 candle lies inside that last half hour, and I test both the momentum and the gamma channel directly. What I add is a single minute within that window, tested out of sample under a pre-registered plan. The holdout design and the luck diagnostics follow the work on data mining and multiple testing in return predictability (Harvey et al., 2016). Pre-registration (Nosek et al., 2018) is still rare in empirical finance, and the paper applies a plan, hash and ledger protocol to a sequence of trading studies and reports the tests that failed.

Outline. Section 2 describes the data and Section 3 the research design. Section 4 tests the signal, and Section 5 studies strike choice and early exits under a pricing model. Sections 6 and 7 price the trade at real quotes and examine how it would have been executed. Sections 8 to 10 report the studies that tried to improve the rule, and Section 11 the re-validation. Sections 12 and 13 read the candle from the index and from SPY, Section 14 searches the other closing candles, and Section 15 tests the closing-auction explanation. Section 16 reports the forward evidence, Section 17 discusses what the studies imply together, and Section 18 concludes.

2 Data

The study combines one-minute bars for the index and for SPY, real option quotes and trades, the exchanges' closing-auction imbalance messages and a broker's own one-minute bars. Table 1 lists the sources.

Table 1: Data sources.

Source	Content	Period	Used for
Dukascopy	US500 CFD one-minute bars, regular trading hours	2013–2026	Signals; index outcomes
Databento OPRA	SPXW best bid and offer at one-minute marks, 15:00–16:01 ET, the 25 strikes nearest the money, calls and puts	2013–2026	Option prices (Sections 6–8)
Databento OPRA	SPXW one-minute trade bars for the same contracts	2013–2026	Execution (Section 7)
Databento OPRA	Daily open interest, all SPX and SPXW options	2013–2026	Gamma proxy (Section 9)
Databento OPRA	XSP and SPY quotes; XSP trade bars	2021–2026	Smaller contracts (Section 10)
Cboe	Official SPX daily close	2013–2026	Option settlement
Robinhood	SPX and SPXW one-minute bars	2026	Model calibration; forward test
CNBC	SPX index one-minute bars (archive)	2021–2026	Index candle (Section 12)
CNBC	SPY one-minute bars (archive; traded prices, not dividend-adjusted)	2021–2026	SPY candle (Section 13)
Databento	NYSE and Nasdaq closing-auction imbalance messages, 15:50–16:00 ET on signal days, for SPY's 50 largest holdings (SEC N-PORT reports)	2021–2026	Imbalance test (Section 15)

Note: The index sample runs from 2013-02-28 to 2026-10-20 and the OPRA SPXW sample from 2013-04-01 to 2026-10-20; the XSP and SPY sample starts on 2021-01-01. Robinhood data begin on 2026-09-21. The CFD series was extended to 2026-11-09 for the comparison in Section 11.

Signals and index outcomes come from Dukascopy's US500 CFD one-minute bars during regular trading hours, restricted to a validated set of full sessions. Discovery uses 2013–2020 (1,861 sessions) and the holdout 2021–2026 (1,446 sessions). The CFD stands in for SPX, and index outcomes are measured to the close of the 15:59 bar rather than to the official settlement. Section 12 reads the signal candle instead from an archive of SPX index one-minute bars, and Section 13 from an archive of SPY one-minute bars. On the 22 days where both are available, the CFD and SPX moves from 15:51 to the close had the same sign on 86% of days.

Real option prices come from Databento's OPRA feed: the consolidated best bid and offer at one-minute marks for SPXW contracts expiring that day, from 15:00 to 16:01 ET, for the 25 strikes within 60 points of the at-the-money strike at 15:51. A quote stamped HH:MM:00 is the quote at the open of that minute. One-minute trade bars for the same contracts support the execution analysis, and daily open interest on every SPX and SPXW option supports the dealer-gamma proxy. Of the 3,287 index days, 2,149 had an SPXW expiry that day; the rest are excluded from every option study. Options settle at intrinsic

value on Cboe's official SPX close, which matched the broker's daily close on all 362 days on which both exist.

Finally, Robinhood's SPX and SPXW bars serve two purposes. Seven days of real one-minute option prices calibrate the pricing model of Section 5, and since 21 October 2026 a read-only collector has saved the day's SPX and SPXW one-minute bars after every close for the forward test (Section 16). The broker keeps option minute bars for only about a day, so a day that is not collected promptly is lost.

3 Research design

3.1 The signal and the trade

Let o, h, l, c be the open, high, low and close of a one-minute bar. The bar has a *strong body* if $h > l$ and

$$|c - o| \geq 0.6\,(h - l), \tag{1}$$

and its direction is the sign of $c - o$. The signal studied throughout is a strong body on the 15:50 bar. Because that bar is complete only at 15:50:59, the trade enters at the open of the 15:51 bar and, in the index tests, is scored on the move to the close of the 15:59 bar.

In the option studies, let S be the index level at entry and $K_{\text{ATM}} = 5\lfloor S/5 + 1/2 \rfloor$ the nearest listed strike. An offset of j points, positive meaning in the money in the trade's direction, buys the call struck at $K_{\text{ATM}} - j$ after an up signal and the put struck at $K_{\text{ATM}} + j$ after a down signal. I consider nine offsets, from 20 points in the money (ITM20) to 20 points out of the money (OTM20) in steps of five. One contract (multiplier 100) is bought per signal and held to cash settlement unless stated otherwise, and each side pays a fee of \$0.09 per contract, with \$0.74 as a sensitivity. I refer to the 15:50 strong-body signal traded at ITM10 as the *current rule*.

3.2 Pre-registration

Each confirmatory study was specified in a written plan before the data it governs was read. The plan's SHA-256 hash and a UTC timestamp were appended to a ledger, and each study's code recomputes the hash at start-up and refuses to read real data unless it matches a ledger entry. Each plan lists what was already known when it was written, and any departure from a plan is reported as a labelled deviation; no result was re-run under a changed rule. Table 2 lists the studies, their primary criteria and their outcomes. Analyses that were not pre-registered are labelled *descriptive* and were not used to decide any test.

3.3 Statistical inference

I report mean P&L per trade and per day, where the daily series takes the trade's P&L on a signal day and zero otherwise. Unless stated otherwise, t-statistics use Newey–West standard errors with five lags on the daily series (Newey and West, 1987). The scan tests its candidates one-sided with a Holm correction (Holm, 1979), and hit-rate intervals are Wilson intervals (Wilson, 1927).

Two studies choose one setting from a large grid, and for them I report a *luck diagnostic*. The signal series is shifted circularly against the outcome days, by every shift from 21 days to $n - 21$ in steps of three. A shift keeps the signal's own structure but breaks its link to the

Table 2: The pre-registered studies.

Study	Section	Registered	Primary criterion	Outcome
Candle scan	4	2026-10-21	Holm-corrected test of each frozen candidate on 2021–2026, and an accuracy bar	One of three candidates passes
Strike ladder	5.1	2026-11-08	Nine strikes ranked under a calibrated model	Deep in the money most robust
Early exits	5.2	2026-11-09	A rule beats holding with $t > 2.64$, robust across model variants	No rule passes
Real option prices	6	2026-11-09	Switch only if a new setting leads by $t \geq 2.0$; confirm with $t \geq 2.0$	Current rule kept and confirmed
Re-tuning	8	2026-11-09	Luck diagnostic $p \leq 0.05$, then $t \geq 2.0$ on 2024–2026	Pick fails the luck diagnostic
Intraday momentum	9	2026-11-09	Slope $t \geq 2.0$; filters $p \leq 0.0125$; gamma $p \leq 0.025$	Nothing supported
XSP and SPY	10	2026-11-09	Positive mean with $t \geq 2.0$ on 2021–2026	Neither confirmed
Index candle	12	2026-11-10	Rule read from the index candle: positive mean per day with $t \geq 2.0$	Criterion not met
SPY candle	13	2026-11-10	Rule read from the SPY candle: positive mean per day with $t \geq 2.0$	Retrospective criterion met
Candle search	14	2026-11-10	Best of 7,128 settings on 2021–2023, tested on 2024–2026 only if not at 15:50 and luck diagnostic $p \leq 0.05$	Pick at 15:50; no test
Closing imbalance	15	2026-11-10	Hit rate higher on days the candle agrees with the 15:50 imbalance: $d > 0$ with $t \geq 2.0$	Not confirmed
Real-time exits	7.4	2026-11-10	An exit rule beats holding in both halves of 2021–2023 and has luck diagnostic $p \leq 0.05$; then $t \geq 2.0$ on 2024–2026	Stopped at the coverage gate
Real-time exits, revised	7.4	2026-11-11	The same, for the 26 rules that read only the option's bid	No eligible rule; no test
Forward options log	16	2026-10-30	Running record of real option outcomes	Four priced trades so far

Note: Registration dates are UTC. The limit-order and fixed-minute sale analyses of Section 7 and the re-validation of Section 11 were not pre-registered.

outcomes. The whole search is repeated for each shift and its best score kept. With S shifts, of which k produce a best score at least as good as the real one, $p = (1 + k)/(1 + S)$. The diagnostic asks whether the real winner beats the winner that unrelated signals would find. It is approximate, because expiry frequency, volatility and pricing change over the sample.

3.4 Option prices

The first option studies had no historical option prices and priced each trade with a Bachelier model (Bachelier, 1900):

$$C = (S - K)\,\Phi(d) + \sigma\,\phi(d), \qquad P = C - (S - K), \qquad d = \frac{S - K}{\sigma}, \tag{2}$$

where σ is the standard deviation, in index points, of the move from entry to the close. Each day's σ is proportional to that day's one-minute realised volatility, scaled to the nine minutes to the close. The constant, and a skew factor for each strike, were calibrated on seven days of real Robinhood SPXW prices (Appendix A). The model charges a $0.10 half-spread on entry. To compare strikes on equal terms, every modelled trade is rescaled to a common index level of 7,709.58 before it is priced, so model results are in *level-normalised* dollars. The later studies replace the model with real OPRA quotes, buying at the ask at the open of the entry minute; their results are in dollars at the index level of the time.

4 Does a closing candle predict the close?

I defined 41 features, each of which outputs up, down or no trade using only bars completed before the entry: the direction of every one-minute candle from 15:30 to 15:50; five candle shapes (strong body, hammer, shooting star, engulfing, big candle) on the 15:50 one-minute candle and on the 15:45–15:49 five-minute candle; five-minute directions; and four whole-day features. Features with $|t| \geq 2.5$ on 2013–2020 qualified, up to eight. The list of qualifying features was frozen and hashed before any 2021 bar was read. On 2021–2026 each candidate faced a one-sided test with a Holm correction at 5%. A candidate that passed was called *accurate* only if the lower bound of its 95% Wilson interval exceeded the hit rate at which buying the at-the-money option breaks even.

Table 3: The candle scan: frozen candidates on the 2021–2026 holdout.

Panel A. The three candidates

Candidate	2013–20	2021–26 (95% CI)	trades	Holm p	predicts	accurate
15:50 strong body	54.1%	**57.5%** (53.6–61.2)	644	6.3×10^{-5}	yes	**yes, just**
15:50 direction, any body	52.2%	54.0% (51.4–56.6)	1,445	0.011	yes	no
Day so far (09:30–15:49)	52.0%	48.4% (45.8–51.0)	1,446	0.53	no	no
P3 reference (not tested)	*51.8%*	*53.6%* (51.1–56.2)	*1,442*	—	—	—

Panel B. The strong-body candle by subperiod (descriptive)

Strong body	trades	hit rate (95% CI)	t	option P&L / trade (t)
2021–2022	216	63.0% (56.3–69.1)	3.56	+$80.7 (2.33)
2023–2026	428	54.7% (49.9–59.3)	2.27	+$28.6 (1.11)

Note: Panel A gives hit rates in discovery (2013–2020) and on the holdout, with 95% Wilson intervals. The strong-body break-even is 49.5% at a $0.09 fee (51.8% size-neutral); with 644 trades, "accurate" requires an observed hit rate of 53.4% (55.7% size-neutral). The P3 row is a reference and was not tested. Panel B prices the at-the-money option with the scan's cost model. By year from 2021 to 2026 the hit rate was 64.9, 61.0, 54.0, 53.8, 57.3 and 53.3%.

Panel A of Table 3 shows that three features qualified. Two of them, both on the 15:50 candle, predicted the 2021–2026 move, but only the strong-body candle also cleared the accuracy bar. It was right on 57.5% of 644 trades. Rescaled to today's index level, these trades moved an average of +1.82 points in the signal's direction over nine minutes. The third candidate, a whole-day feature, failed out of sample.

I ran two checks, one against chance and one against a generic intraday pattern. When the 41 features were shifted out of line with the 2013–2020 outcomes and the scan was re-

run 304 times, about 1% of runs found as many qualifying features as the real scan, and none found one as strong as the best real feature. The same rules moved four hours earlier, entering at 11:51, showed nothing in either period.

The result is not entirely new. Both passing features sit on the 15:50 candle and agree with the direction of the move from 15:50 to 15:51 (an effect I call P3, found in the project's earlier exploratory work) on 98% and 93% of days. I knew how P3 had behaved over 2013–2026 before running the scan, so the holdout was not blind for these features. I therefore read the result as a known effect passing a stricter test, not as a new discovery. The strong body does matter, because on 15:50 candles without one, trading the direction earned nothing.

The effect has also weakened since 2023. Panel B of Table 3 shows that most of the edge comes from 2021–2022. In 2023–2026 the hit-rate interval includes 50%, and the modelled option P&L is not significant. At that effect size, a test with 80% power would need roughly 19 years of signals, so no single year can settle the question.

5 Strike choice and early exits under a pricing model

Before real option quotes were available, two pre-registered studies used the calibrated model of Section 3.4 to price the 644 holdout trades. The first asks which strike best expresses the signal, and the second whether a stop-loss or take-profit improves on holding to settlement. Both report a headline calibration, variant (a), and alternatives that change the volatility, the costs or the skew (Appendix A). Under the model, strikes deep in the money were the most robust and no early exit improved on holding.

5.1 Which strike

Table 4: The strike ladder under the calibrated model.

Strike	model price	mean P&L / contract	t	profitable	mean return	median return
ITM20	\$2,036	+162.21	3.89	56%	+8%	+6%
ITM15	\$1,580	+142.62	3.50	55%	+9%	+6%
itm10	**\$1,098**	**+166.08**	**4.25**	**54%**	**+15%**	**+8%**
ITM5	\$693	+160.97	4.49	51%	+23%	+3%
ATM	\$367	+154.63	4.96	42%	+45%	−50%
OTM5	\$205	+92.10	3.59	27%	+53%	−100%
OTM10	\$204	−34.34	−1.47	14%	+7%	−100%
OTM15	\$120	−21.91	−1.11	9%	+13%	−100%
OTM20	\$1,683	−1,628.66	−3.62	1%	−74%	−100%

Note: Variant (a), 644 modelled strong-body trades on 2021–2026, \$0.09 fee, level-normalised dollars per contract. A model, not observed P&L. The OTM20 row (grey) is an artefact of the skew rule; the OTM10 and OTM15 rows inherit skew factors of 2.09 and 2.26 and are unreliable for the same reason. Table A2 reports every variant.

Table 4 shows that strikes 10 to 20 points in the money were the most robust. Their mean stayed between +\$102 and +\$214 per contract in every credible variant, a majority of trades were profitable, and the typical trade made money. Deep in-the-money options carry little time value, so they keep nearly all of a small move. At-the-money options earn a higher return on premium, but from fewer, larger winners. The median at-the-money trade lost about half its premium, and the mean depends heavily on the assumed price of volatility. The ranking held in both subperiods, although the level fell after 2022.

The pre-registered skew factor, the median ratio of market to model price, failed for far out-of-the-money strikes. On the calm calibration days the normal model priced a 20-point out-of-the-money option at almost nothing, while the market charged at least the minimum tick, so the factor came out at 54. Applied to volatile days, it produced absurd prices. Because the rule was fixed in advance, its output is reported unchanged and marked as unreliable. The real prices of Section 6 settle the question for those strikes.

5.2 Early exits

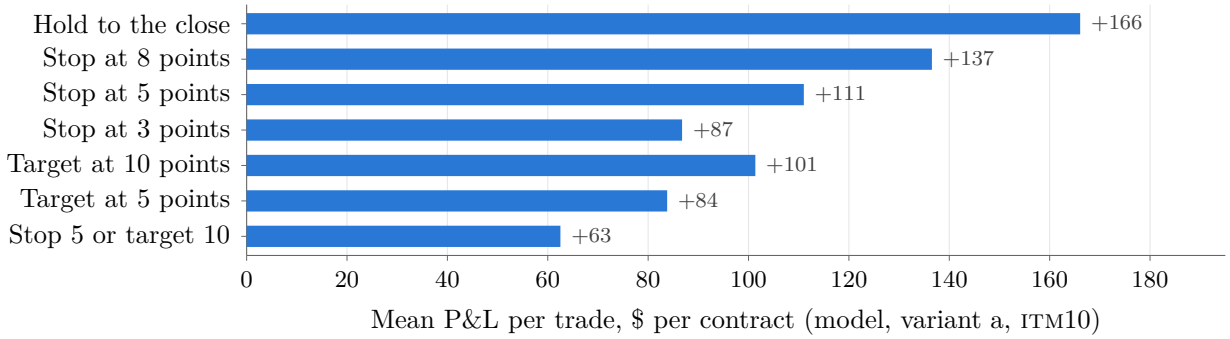
Six exit rules watch the index's one-minute closes from 15:51 to 15:58: stops at 3, 5 or 8 points against the trade (S3, S5, S8), take-profits at 5 or 10 points in its favour (T5, T10), and a stop at 5 combined with a target at 10 (B5_10). A rule that fires at the close of a minute sells at the next minute's open, paying the half-spread and a second fee. The baseline (H) holds to settlement. A rule counts as better than holding only if, at ITM10 in variant (a), its mean paired difference from holding is positive with $t > 2.64$ (two-sided 5%, Bonferroni over six rules), and the difference is also positive in two alternative variants. The plan records that ITM10 was chosen as the headline strike after the strike results had been seen.

Table 5: Early exits under the calibrated model.

Rule	sold early	mean	t	profitable	5th pct	worst	vs hold (t)
H (hold)	0%	+166.08	4.25	54%	−1,186	−3,601	—
S3	51%	+86.76	2.65	38%	−695	−1,192	−79.32 (−3.35)
S5	35%	+111.02	3.14	46%	−823	−1,481	−55.06 (−2.80)
S8	20%	+136.54	3.45	50%	−1,037	−1,481	−29.54 (−2.15)
T5	44%	+83.79	3.23	63%	−1,120	−3,601	−82.29 (−2.78)
T10	19%	+101.36	3.39	56%	−1,186	−3,601	−64.72 (−2.89)
B5_10	51%	+62.52	2.34	48%	−810	−1,481	−103.56 (−3.79)

Note: ITM10, variant (a), 644 modelled trades, level-normalised dollars per contract, $0.09 fee. "Sold early" is the share of trades the rule closes before settlement. The last column is the mean paired difference from holding, with its Newey–West t.

Figure 1: Early exits against holding to the close.



Note: 644 modelled trades (2021–2026), $0.09 fee. Every stop and take-profit rule earned less on average than holding; stops bought a smaller worst case (Table 5).

Table 5 and Figure 1 show that no rule beat holding. Every rule's mean difference from holding was negative, at every strike reported and in every variant. Stops cut the worst loss

by more than half, but they cost money on average and reduced the share of profitable trades. The likely explanation is that the signal is a small drift over nine minutes, so many adverse moves reverse before the close, and every early sale pays the spread again. Take-profits raised the share of winning trades and lowered the mean, because they cut the winners short and do nothing for the losers. Sections 7.3 and 7.4 revisit early exits at real prices.

6 Real option prices

The model studies leave two questions open, whether the edge is still there at the prices actually quoted, and whether the current rule is the best nearby setting. A pre-registered study addressed both with real OPRA quotes, tuning on 2013–2020 and confirming once on 2021–2026.

6.1 Design

At the entry minute, the spot S is implied by put–call parity: it is the median of $K + C_{\text{mid}} - P_{\text{mid}}$ over the three strikes where the call and put mids are closest. Strikes then follow Section 3.1. The trade buys at the ask at the open of the entry minute and settles on the official close, so its P&L per contract is

$$\text{P\&L} = 100\,\big(\text{intrinsic value at the official close} - \text{ask}\big) - \text{fee}. \tag{3}$$

A trade with no implied spot or no valid quote is dropped from that setting and counted.

The search grid combines the candle minute (any one-minute candle from 15:40 to 15:50, entering at the next minute's open), the body threshold (0.40 to 0.90 in steps of 0.05), the direction (with or against the candle) and the strike (ITM20 to OTM20), for 2,178 settings in all. A setting's score on 2013–2020 is its mean P&L per day averaged with its neighbours at adjacent thresholds, which favours a plateau over a sharp peak. A setting is eligible with at least 150 trades, at least 50 in each half of the period, and a positive mean in both halves. The best eligible setting would replace the current rule only if its luck diagnostic gave $p \le 0.05$ and its per-day lead over the current rule had $t \ge 2.0$. The chosen rule is confirmed on 2021–2026 if its mean P&L per day is positive with $t \ge 2.0$. The plan states that the 2021–2026 index outcomes had already been used to test the signal, so the confirmation is fresh for option prices but not for the index moves.

6.2 Tuning on 2013–2020

Table 6: Tuning at real prices, 2013–2020: the eligible settings.

Cell (candle, body, direction, strike)	score	mean	trades	2013–16	2017–20	eligible
15:42, ≥ 0.60, with, ATM	+25.61	+26.24	250	+0.29	+34.29	yes
15:50, ≥ 0.40, with, ATM	+4.62	+5.34	302	+0.02	+6.77	yes
Current: 15:50, ≥ 0.60, with, ITM10	−12.65	−7.70	216	−21.57	−3.82	no

Note: 2013-04-01 to 2020-12-31, real asks, \$0.09 fee, dollars per contract per day. Only these two of the 2,178 settings were eligible. Luck diagnostic: 262 shifts, 142 of them with no eligible setting (counted as the worst); 95th percentile of the shifted best score +\$21.92; $p = 0.034$. Best minus current: +\$40.17 a day, Newey–West $t = 1.56$ over 593 days.

Tuning on 2013–2020 had little to choose from, because only two of the 2,178 settings were eligible (Table 6). The better one, the 15:42 candle with a 60% body at the money, passed the luck diagnostic, but its lead over the current rule was not significant, so the current rule went forward unchanged. The current rule itself lost money at the ask in this period. Many of its signal days could not be priced at all, mostly because too few strikes had valid quotes to imply a spot. In the confirmation period only one signal day could not be priced.

6.3 Confirmation on 2021–2026

Table 7: Confirmation: the current rule at real asks.

Period	days	$/day	t	trades	$/trade	won	median	worst
2021–2026	**1,321**	**+29.87**	**2.40**	**585**	**+67.46**	**54%**	**+44.91**	**−2,210**
2021–2022	376	+48.21	2.01	157	+115.46	57%	+65.91	−1,361
2023–2026	945	+22.58	1.56	428	+49.85	53%	+35.91	−2,210

Note: The current rule (15:50 candle, body at least 60%, with the candle, ITM10, held to the official close) at real asks, $0.09 fee, dollars per contract. t is Newey–West on the per-day series. Subperiods are descriptive. Mean per trade by year: 2021 +$175 (67 trades), 2022 +$71 (90), 2023 +$4 (113), 2024 +$20 (106), 2025 +$65 (117), 2026 +$121 (92).

Table 7 shows that the current rule was confirmed. It earned +$29.87 a day with $t = 2.40$, against a bar of 2.0. The mean trade made +$67.46, slightly more than half of all trades were profitable, and the worst trade lost $2,210 on one contract. Figure 2 shows the full record. This confirmation has three weaknesses. It confirms the current rule at real prices but does not show that it is the best setting, since no other setting was evaluated on this period (Section 8 returns to that question). I had already seen the index outcomes. And the edge is weaker since 2023, when it is not significant on its own, so much of the result comes from 2021–2022.

6.4 The role of the bid-ask spread

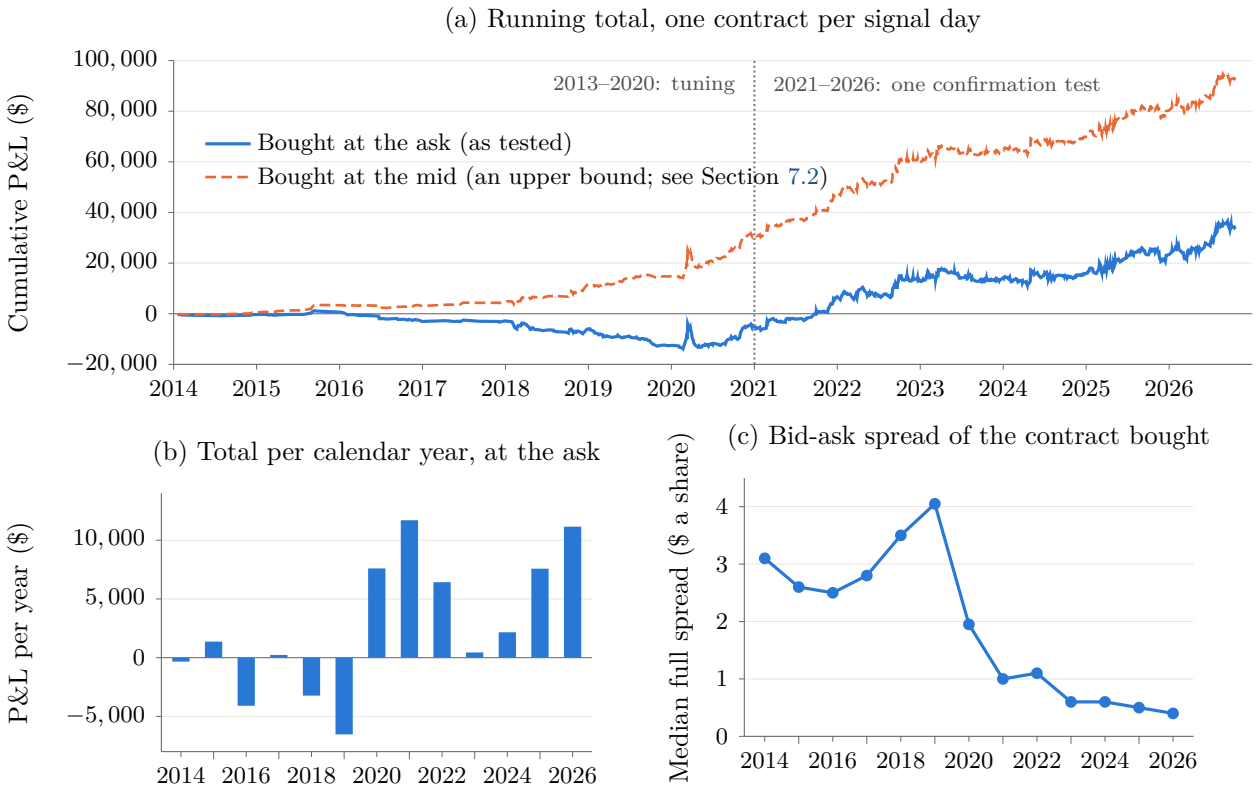
Table 8: The current rule bought at the ask and at the mid.

Period	bought at	trades	$/day	t	$/trade	won	median full spread
2013–2020	ask	216	−7.70	−0.47	−23.14	42%	$2.90
	mid	216	+48.19	2.87	+144.78	56%	
2021–2026	ask	585	+29.87	2.40	+67.46	54%	$0.60
	mid	585	+46.90	3.71	+105.90	56%	

Note: ITM10, $0.09 fee, dollars per contract. Rows at the mid are descriptive and decide nothing. The spread is ask minus bid per share at entry, the median over the priced trades. With a $0.74 fee the ask rows become −$23.79 and +$66.81 a trade ($t = 2.37$ in 2021–2026).

Table 8 and Figure 2(c) show how much the result depends on the spread. At the mid, the same trades made money in both periods. At the ask they lost money in 2013–2020, when the median quoted spread was about five times its 2021–2026 level, and made money afterwards. Even since 2021, paying the ask instead of the mid costs about $39 a trade, more than a third of the edge at the mid. How the order is filled therefore matters a great deal, and Section 7 looks at it.

Figure 2: The current rule at real option prices.



Note: The current rule (15:50 candle, body at least 60%, with the candle, ITM10, held to the official close) at real Databento OPRA prices, $0.09 fee, dollars per contract at each year's prices. (a) The 801 priced signal days, 2014–2026. At the ask the rule lost $4,997 over the 216 trades of 2013–2020 and made $39,463 over the 585 of 2021–2026. The dashed line buys at the mid and is only an upper bound: a limit order at the mid made about what paying the ask made, because it missed the best days (Section 7.2). (b) 2023 and 2024 were close to zero. (c) Median ask minus bid at entry. The narrowing spread is why the same signal lost money at the ask before 2021 and made money after it. Before 2016 SPXW had a same-day expiry only on some days, so the early years have few trades. Sources: Databento OPRA; Cboe; Dukascopy.

6.5 Strikes at real prices

Table 9 and Figure 3 show that at real prices the strikes rank as they did in the model. Strikes 10 to 20 points in the money earned the most and won most often, and ITM10 needs about half the premium of ITM20. At the money the mean was positive but most trades lost. Ten or more points out of the money, the option rarely paid.

6.6 How accurate was the model?

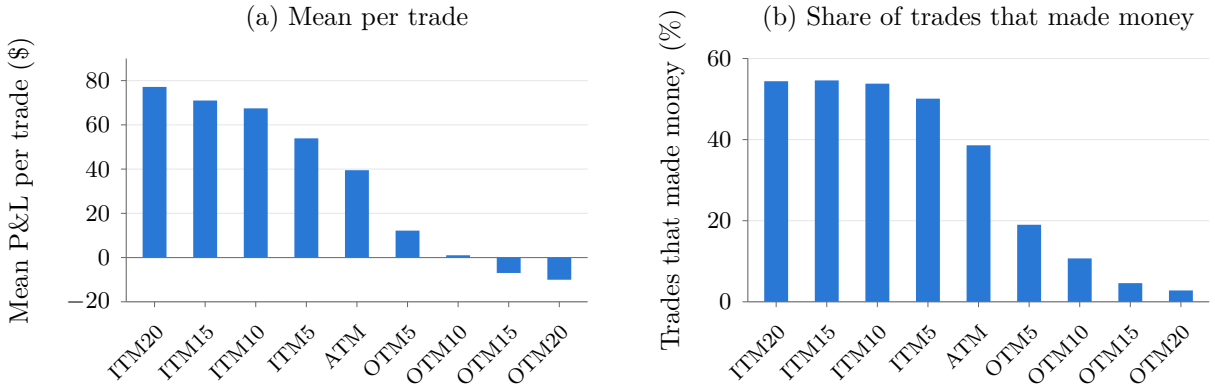
Table 10 sets the model against the real trade on the same days, contracts and index levels. On these terms the model implied +$100.70 a trade at ITM10 and the real trade made +$67.77, a significant shortfall. (The strike ladder's larger figures are in level-normalised dollars.) In the money the model priced the option well, with the real mid within about 2% of the model price, but the real ask sat further above the mid than the model's flat half-spread allowed. At the money the model was too cheap, which is why its at-the-money results looked so much better than the real ones. Far out of the money its skew factors made it unusable. The settlement price was not the cause, since the official close differed from the

Table 9: The current rule's signal at every strike, real asks, 2021–2026.

Strike	trades	$/trade	won	median	worst	$/day	t
ITM20	586	+77.15	54%	+51.91	−2,670	+34.20	2.61
ITM15	584	+71.00	55%	+52.41	−2,330	+31.41	2.43
itm10	**585**	**+67.46**	**54%**	**+44.91**	**−2,210**	**+29.87**	**2.40**
ITM5	585	+53.90	50%	+2.91	−1,900	+23.87	2.07
ATM	583	+39.50	39%	−110.09	−1,640	+17.46	1.87
OTM5	585	+12.18	19%	−65.09	−1,390	+5.40	0.78
OTM10	581	+1.04	11%	−25.09	−1,180	+0.46	0.10
OTM15	566	−6.96	5%	−15.09	−1,000	−3.02	−1.03
OTM20	529	−10.02	3%	−10.09	−840	−4.19	−2.34

Note: $0.09 fee, dollars per contract. Descriptive, not a second test: choosing a strike from this table would be choosing on the confirmation period.

Figure 3: The current rule's signal at every strike, real asks, 2021–2026.



Note: $0.09 fee, dollars per contract. In-the-money strikes earned the most and won most often; out of the money the option rarely paid. Descriptive, not a second test (Table 9). Sources: Databento OPRA; Cboe; Dukascopy.

CFD's 15:59 close by 1.33 points on average, with almost no bias.

7 Execution

The real-price results assume a fill at the quoted ask at 15:51:00. Here I ask whether that is realistic, whether a cheaper order would have done better, and what a late entry or an early sale costs. Apart from the real-time exit study of Section 7.4, which was pre-registered, all of it is descriptive. The limit-order and early-sale analyses were not pre-registered, and 2021–2026 had already confirmed the rule. In brief, since 2021 the contract traded near the ask on almost every signal day, neither limit orders at the mid nor sales before the close improved on paying the ask and holding to settlement, and later entries earned less.

7.1 Did the contract trade near the ask?

The ask is only a quoted price, and I do not observe what an order would actually have paid. OPRA trade bars show whether the contract traded in the entry minute and at what prices, and Table 11 summarises them. Since 2021 it traded in the entry minute on almost every signal day, against just over half the days before, and on nearly all of those days the minute's

Table 10: The pricing model against real prices on the same trades.

		price per share			ask / model	P&L per contract		
Strike	n	model	real mid	real ask	mean / median	model	real	real − model (t)
ITM20	586	20.09	20.07	20.58	1.02 / 1.02	+94.34	+77.16	−17.18 (−2.01)
ITM15	584	15.37	15.22	15.68	1.02 / 1.01	+73.45	+71.82	−1.63 (−0.18)
itm10	**585**	**10.29**	**10.51**	**10.89**	**1.06 / 1.05**	**+100.70**	**+67.77**	**−32.93 (−3.87)**
ITM5	585	5.85	6.28	6.51	1.12 / 1.11	+96.90	+51.57	−45.33 (−5.63)
ATM	583	2.40	3.08	3.18	1.50 / 1.33	+92.67	+35.15	−57.52 (−7.37)
OTM5	585	0.93	1.37	1.44	— / 1.87	+45.59	+9.32	−36.27 (−5.52)
OTM10	581	0.69	0.66	0.71	— / 2.65	−9.04	+2.31	+11.35 (1.27)
OTM15	564	0.34	0.37	0.41	— / 15.31	−13.42	−7.07	+6.35 (0.84)
OTM20	524	4.77	0.23	0.27	— / 10.82	−470.67	−10.83	+459.84 (1.79)

Note: The strike ladder's model (variant (a)) against real prices on the 644 strong-body trades of 2021–2026, at the index levels of the time (not rescaled), on the days with a 15:51 quote for that contract. The model settles on the CFD's 15:59 close with a $0.10 half-spread; the real trade buys at the ask and settles on the official close; both pay $0.09. Out of the money the mean ratio is not meaningful, because model prices can be close to zero, so only the median is shown.

Table 11: Trading in the entry minute.

Current rule, ITM10, entry minute 15:51	2013–2020	2021–2026
Priced entries (all with a trade-bar file)	216	585
Traded in the entry minute	122 (56%)	580 (99%)
median contracts traded in the minute	10	75
lowest trade at or below the 15:51 ask	95%	95%
lowest trade at or below the 15:51 mid	52%	83%
minute's last trade minus the mid, mean	+$0.58	+$0.16
minute's last trade minus the mid, median	+$0.57	+$0.25

Note: From OPRA one-minute trade bars. Prices are per share ($100 a contract per $1). The indented rows are shares or values over the entries that traded in the minute.

lowest trade was at or below the 15:51 ask. The minute's last trade sat above the mid but inside the half-spread, and often above the ask, as one would expect when the price keeps moving in the signal's direction. A trade bar does not show the size available, the queue, or what a particular order would have paid.

7.2 Limit orders and late entry

Since the trade made far more at the mid than at the ask (Table 8), the obvious thing to try is a limit order. I simulated six order styles, fixed before any result was computed: paying the 15:51 ask (P0); a limit at the 15:51 mid for one minute, else no trade (P1); the same, but counting a trade at the limit as a fill (P1t); P1 followed by paying the 15:52 ask if unfilled (P2); a mid limit left working for three minutes (P3); and a limit halfway between the mid and the ask, then the 15:52 ask (P4). A limit counts as filled if a trade printed strictly below it during a minute it was working, or if the quote at the end of that minute offered the option at or below it. These are proxies and not estimates of execution. Trade prints cannot show whether a resting order would have executed, the order is assumed to be live with no delay, and one contract is assumed.

No order style beat paying the ask (Table 12). On 2021–2026 every style earned roughly

Table 12: Order styles for the current rule, 2021–2026.

	2021–2026, 585 priced signal days				minus P0, $/day	
Order style	filled	$/day	t	minus P0 (t)	2021–22	2023–26
P0 pay the 15:51 ask	585 (100%)	+67.46	2.29	—	—	—
P1 mid, 1 minute, else skip	489 (84%)	+64.57	2.37	−2.89 (−0.24)	−22.31	+4.23
P1t as P1, touch counts	501 (86%)	+61.04	2.23	−6.42 (−0.56)	−22.31	−0.59
P2 mid, 1 minute, then 15:52 ask	585 (100%)	+69.63	2.38	+2.17 (0.55)	+13.06	−1.82
P3 mid, 3 minutes, else skip	534 (91%)	+64.89	2.31	−2.57 (−0.26)	+6.44	−5.87
P4 halfway, 1 minute, then 15:52 ask	585 (100%)	+62.23	2.13	−5.23 (−1.63)	−2.36	−6.29

Note: ITM10, $0.09 fee, dollars per contract per priced signal day; an unfilled order counts as $0. The subperiods have 157 (2021–2022) and 428 (2023–2026) signal days. The t-statistics here treat signal days as independent, so P0's t is 2.29 rather than the per-day Newey–West 2.40 of Table 7.

Table 13: Why the mid limit does not help, and what a late entry costs.

(a) The same entries at the ask, split by whether P1's mid limit filled

Period	P1	days	$/trade at the ask ($t$)	won
2021–2026	filled	489	+42.52 (1.31)	52%
	not filled	**96**	**+194.49 (2.80)**	**61%**
2013–2020	filled	69	+44.11 (0.37)	46%
	not filled	147	−54.70 (−1.42)	40%

(b) The same contract bought at a later minute's ask, $/trade

Buy at	2013–2020	2021–2026 (t)	2021–2022	2023–2026
15:51	−23.14	+67.46 (2.29)	+115.46	+49.85
15:52	−68.04	+55.75 (2.02)	+111.07	+35.46
15:53	−58.09	+57.89 (2.16)	+128.30	+32.06
15:55	−128.97	+23.01 (1.11)	+51.10	+12.71

Note: Current rule, ITM10, $0.09 fee, dollars per contract. Panel (b) uses the days with a quote at every one of these minutes: 216 (2013–2020), 585 (2021–2026), 157 (2021–2022) and 428 (2023–2026), which is every priced day.

what paying the ask earned, and every difference was within noise. Panel (a) of Table 13 suggests that the cause is adverse selection. A limit at the mid fills when someone sells below the mid, which happens most when the option is getting cheaper, that is, when the trade is going wrong. When the trade works at once, the price moves away and the limit is left unfilled. The days on which the mid limit filled earned well below the average at the ask, and the days it missed earned far more. The limit saves money on the days it fills and misses the best days, and the two effects roughly cancel. Before 2021, with spreads about five times wider, the mid limit did help, but on a small sample.

Entering late also costs money. Bought at a later minute's ask, the same contract earned less the later the entry (Table 13, panel b), about $12 a trade less at 15:52 and about two-thirds less at 15:55. The rule's results assume entry at 15:51:00 exactly.

7.3 Selling before the close at real bids

As a check on the model's exit results, each of the 585 confirmation trades, bought at the 15:51 ask, was also sold at the bid at the start of each later minute, paying the fee twice. Every fixed exit minute from 15:52 to 15:59 earned less than holding to settlement, by $79 to $136 a trade, and no year favoured an early exit. Selling at the best minute in hindsight would have earned far more, which shows only how much the price moves after entry.

7.4 Real-time exit rules at real bids

A fixed exit minute ignores what the price does after entry. A pre-registered study tested rules that watch the option's own bid and sell when it moves. Each of the rule's 270 entries of 2021–2023, bought at the 15:51 ask at ITM10, was followed through one-minute quote snapshots to 15:59. A rule decides at each minute from 15:52 to 15:58 and sells at the next minute's bid, paying a second fee; a trade it never sells settles. Twenty-six rules were fixed in advance: trailing stops that sell when the bid falls 10, 20, 30, 40 or 50% below its highest point since entry (5 rules); profit locks that, once the bid has risen 10, 20 or 30% above the price paid, sell if it gives back a third, two-thirds or all of that best gain (9); and take-profits that sell once the bid is 10, 20 or 30% above the price paid, from 15:52, 15:54, 15:56 or 15:58 (12). A rule was eligible if it sold early on at least 25 trades, at least 10 in each half of the tuning period (January 2021 to September 2022 and October 2022 to December 2023), and beat holding on average in both halves. The eligible rule with the best score, its mean improvement on holding averaged with those of its neighbours in the grid, would then have faced a block sign-flip luck diagnostic ($p \leq 0.05$) and a single test on 2024–2026, passing if its mean improvement was positive with Newey–West $t \geq 2.0$.

The study ran under two plans. The first (locked 2026-11-10 22:17 UTC) also included 18 rules that trail the index level implied by put–call parity. It required complete quotes for at least 95% of the entries and stopped at that gate before any rule was computed: only 197 of the 270 entries were complete, because the implied index level, which needs three strikes with a bid on both sides, was often missing in the last minutes (on 70 entries at 15:58 alone). A revised plan (locked 2026-11-11 00:01 UTC), written after that stop and its counts were known, kept only the 26 rules that read the option's own bid and required complete option quotes, which all 270 entries had. It was run once.

Table 14 reports the outcome. No rule was eligible, so neither the luck diagnostic nor the test was run. Every one of the 26 rules earned less than holding on 2021–2023, by $9 to $120 a trade. Take-profits did worst. Selling once the bid was 10 to 30% above the price paid cost $49 to $107 a trade against holding, with Newey–West t between -3.2 and -5.6. They sell the trades that would have kept rising and do nothing for the losers, and every early sale gets the bid while holding settles at the index value with no spread. I looked at 2024–2026 after the decision and it agrees: take-profits trailed holding by $39 to $108 a trade, and the trailing stops and profit locks were within $31 of holding, none with $|t|$ above 1.6. Selling at the best minute in hindsight would have added $224 a trade on 2021–2023, which again measures how much the price moves and is not a gain any rule here could capture.

Table 14: Real-time exit rules at real bids.

		2021–2023 (tuning)		2024–2026
Family	Rules	Against holding	Least bad	Against holding
Trailing stop	5	−120.44 to −13.79	TS50	−30.84 to +3.97
Profit lock	9	−37.86 to −8.94	PL30_100	−26.96 to +8.62
Take-profit	12	−106.88 to −49.12	TP1558_30	−108.42 to −39.28
Sale at a fixed minute	8	−168.26 to −111.74	15:57	−120.04 to −51.12

Note: ITM10, bought at the 15:51 ask; dollars per contract. "Against holding" is the range, across the family's rules, of the mean paired difference from holding to settlement over all trades, counting zero where a rule held. Holding earned +$68.78 a trade on 2021–2023 (270 trades) and +$66.32 on 2024–2026 (315). TS50 sells when the bid falls 50% below its peak; PL30_100, once the bid has been 30% above the price paid, sells if it falls back to that price; TP1558_30 sells from 15:58 once the bid is 30% above the price paid. The 2024–2026 column was computed after the decision and decides nothing; the fixed-minute row is descriptive in both periods.

8 Re-tuning on recent data

Tuning on 2013–2020 could not tell nearby thresholds or candle minutes apart, because spreads were wide, many signal days could not be priced, and only two settings were eligible. A further pre-registered study repeated the search on 2021–2023, the first three years with tight spreads and, from 2022, daily SPXW expiries. It would test its pick once on 2024–2026. The pick failed the luck diagnostic, so nothing was tested and the current rule was kept.

Pricing and the grid of 2,178 settings are as in Section 6. A setting is eligible with at least 100 trades, at least 40 in each half of 2021–2023, a positive mean in each half, and at least 95% of its signal days priced. The pick goes to the test only if it lies outside the current rule's family (the 15:50 candle, a 60% body and the candle's direction, at any strike), its luck diagnostic gives $p \leq 0.05$, and its paired difference from the current rule on 2021–2023 is positive. On 2024–2026 the pick would be adopted only if both its mean per day and its paired lead over the current rule were positive with $t \geq 2.0$. Tuning and testing were separate runs: the tuning step could not read any date after 2023, and it recorded its decision, with a hash, before the test step could run.

Table 15: The re-tuning pick and the current rule.

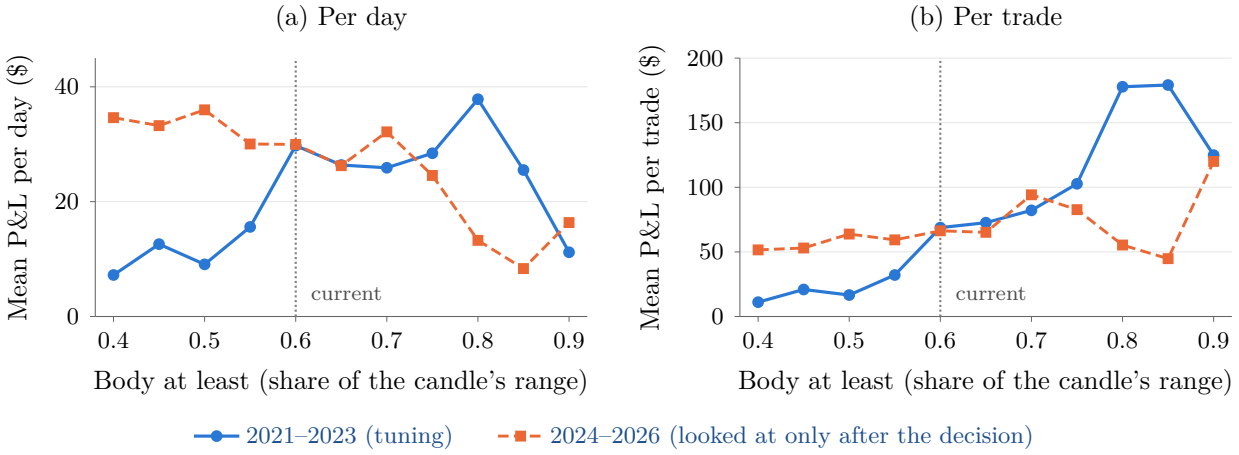
Rule	period	days	$/day	t	trades	$/trade	won	worst
Pick: 15:50, ≥ 0.80, with, ITM20	**2021–2023**	**625**	**+39.70**	**2.94**	**133**	**+186.55**	**62%**	**−1,687**
	H1	315	+75.61	3.60	70	+340.22	66%	−1,687
	H2	310	+3.21	0.21	63	+15.80	59%	−1,433
Current: 15:50, ≥ 0.60, with, ITM10	2021–2023	624	+29.76	1.78	270	+68.78	55%	−1,361
	H1	314	+63.52	2.31	133	+149.97	56%	−1,361
	H2	310	−4.43	−0.25	137	−10.03	54%	−1,340
2024-01-01 to 2026-10-20, computed after the decision was written (descriptive)								
Pick	2024–2026	697	+18.88	1.30	167	+78.80	51%	−2,670
Current	2024–2026	697	+29.97	1.64	315	+66.32	53%	−2,210

Note: Real asks, official close, $0.09 fee, dollars per contract. t is Newey–West on the per-day series. H1 is 2021-01-01 to 2022-09-30 and H2 is 2022-10-01 to 2023-12-31. The pick failed the luck diagnostic, so the 2024–2026 rows decide nothing; the current rule's results there were already known.

Table 15 reports the outcome. On 2021–2023, 119 settings were eligible. The best was the 15:50 candle with a body of at least 80%, traded with the candle 20 points in the money. It failed the luck diagnostic ($p = 0.21$): about one shift in five found a best score at least as good with signals unrelated to the days. No candidate advanced, the test on 2024–2026 was not run, and the current rule was kept. The pick's tuning record came almost entirely from the first half of the period.

Computed only after the decision was recorded, and deciding nothing, the pick earned less than the current rule on 2024–2026, as one would expect of a winner chosen by luck. Figure 4 shows why the threshold is hard to tune. On 2021–2023 a larger body earned more per trade on far fewer trades, but the pattern did not persist; per day, the current 60% threshold earned about the same in both periods, and no threshold beat it in both. Among the other candle minutes, the best eligible 15:48 setting was steadier than the pick across the two halves, but it is one observation from a large search and remains untested. Even with tight spreads, three years of data cannot tell the nearby settings apart, and choosing the best past result would most likely choose a lucky one.

Figure 4: The body threshold.



Note: 15:50 candle, with the candle, ITM10, real asks, $0.09 fee, dollars per contract. On 2021–2023 a bigger body earned more per trade, on far fewer trades (407 at 0.40, 56 at 0.90). The pattern did not hold on 2024–2026. Per day, 0.60 made about $30 in both periods; the 2021–2023 favourite, 0.80, made $13 a day afterwards (Table B1). Sources: Databento OPRA; Cboe; Dukascopy.

9 Intraday momentum and dealer gamma

Gao et al. (2018) find that the S&P 500's first half-hour return, measured from the previous close, predicts its last half hour on SPY from 1993 to 2013. Baltussen et al. (2021) find that the return from the previous close to 15:30 does the same across futures markets from 1974 to 2020, and attribute the effect to option dealers hedging when they are net short gamma. The 15:50 candle sits inside that last half hour, so momentum might improve the candle trade, or the candle might be momentum in another form. A pre-registered study asked whether the momentum appears in my data (Part A), whether it improves the candle trade (Part B), and whether either is stronger when dealers are short gamma (Part C).

With PC the previous session's close and prices taken at bar opens, the predictor is $r_{\text{rod}} = \ln(P_{15:30}/\text{PC})$ and the target is $r_{\text{last}} = \ln(X/P_{15:30})$, where X is the close of the 15:59

bar. Part A regresses r_{last} on r_{rod} over 2013–2026 and requires a positive slope with $t \geq 2.0$. It is an adapted replication: it uses log returns on a CFD rather than simple returns on SPY or futures, and only 2021–2026 lies outside both papers' samples. Part B tests four filters that keep the candle trade only when it agrees with the day's earlier move. A filter must improve the trade with $p \leq 0.0125$ on 2013–2020, and also in both halves and in a version that uses index returns instead of option prices. Part C builds a net-gamma proxy from the open interest of every SPX and SPXW option, following Baltussen et al. (2021) and assuming that dealers are long every call and short every put. It asks whether momentum (C1) or the candle trade (C2) is stronger when the proxy is negative, each with $p \leq 0.025$.

Table 16: Part A: the momentum regression.

Regression of r_{last} on	sample	n	slope $\times 100$	t	R^2
r_{rod} **(primary)**	**2013–2026**	**3,238**	**+1.567**	**1.02**	**0.0030**
r_{rod}	2013–2020	1,802	+2.866	1.16	0.0086
r_{rod}	2021–2026	1,436	−0.173	−0.18	0.0000
r_{rod}, simple returns	2013–2026	3,238	+1.587	1.04	0.0030
r_{rod}, winsorised 1%/99%	2013–2026	3,238	+1.189	1.08	0.0013
r_{first} (first half hour)	2013–2026	3,236	+1.517	0.57	0.0014
r_{12} (15:00–15:30)	2013–2026	3,204	+9.510	1.30	0.0043
BDLM, E-mini futures	*to May 2020*		*about 6.2*	*about 5.0*	*about 0.034*

Note: An adapted replication on the CFD, 2013-04-01 to 2026-10-20. Only the first row is a test; the rest are descriptive. BDLM is Baltussen et al. (2021). By single year, the t of the r_{rod} slope was 2.12, 1.75 and 1.84 in 2013–2015, and between −1.85 and +0.96 in every year since.

The momentum did not replicate (Table 16). The slope was positive but insignificant over 2013–2026, mildly positive only in 2013–2015, and about zero on 2021–2026, the only period outside both papers' samples. This does not show that the published results are wrong, since samples, instruments and periods differ. It shows only that the effect is not visible in these data since 2013. No momentum filter improved the candle trade either (Table B2, Appendix B): trades that agreed with the day's move did better per trade, but the trades a filter skipped were still profitable on average, so skipping them lowered the total. Neither gamma test passed either (Table B3, Appendix B). Under model prices the candle trade did better on negative-gamma days, but the difference missed its bar ($p = 0.029$ against 0.025), came mostly from 2013–2020, and was absent at real asks on 2021–2026. The proxy therefore gives no reason to skip or add trades. Its positioning is assumed and not observed, and open interest misses positions opened and closed on the same day, which is most 0DTE trading since 2022.

10 Smaller contracts: XSP and SPY

The current rule's SPX contract cost about $1,100 on 2021–2026, more than a small account holds. Two underlyings have 0DTE options about a tenth the size: XSP (Mini-SPX, cash-settled, one tenth of SPX) and SPY (the ETF, settled in shares). A pre-registered study traded the same signal on both at real quotes. On each, the spot is implied from that root's own 15:51 quotes, and the contract is $1 in the money in the candle's direction (about 10 SPX points), bought at the 15:51 ask. XSP is held to the official SPX close divided by ten. SPY, which settles in shares, is sold at the 15:59 bid. Each is confirmed only if its mean P&L

per day is positive with $t \geq 2.0$ on 2021–2026. Because the SPX outcomes on these days were already known, the study measures the cost of trading the signal on smaller contracts, not whether the signal works.

Table 17: XSP and SPY at real quotes, 2021–2026.

Contract	Days	$/day	t	Trades	$/trade	Won	Per $1
XSP ITM1, held	1,025	−15.16	−5.48	313	−49.66	35%	−0.285
SPY ITM1, sold 15:59 bid	1,262	+0.23	+0.23	560	+0.51	45%	+0.004
SPX ITM10, held	1,321	+29.87	+2.40	585	+67.46	54%	+0.061

Note: One contract per signal. "Per $1" is total P&L over total premium paid. The SPX row rebuilds the confirmation of Table 7 from its trade list.

Neither was confirmed (Table 17). XSP lost money reliably and SPY broke even, and the reason in both cases is cost. The SPX trade's edge was about six cents per dollar of premium, and both smaller contracts cost more than that to trade. XSP's median quoted spread was 31% of the option's price, against 5% for SPX on the same days. It has narrowed since 2021 and XSP's results improved with it, but even in 2026 the trade only about broke even. XSP is also thin: on almost half of the signal days too few strikes had two-sided quotes to imply a spot. SPY's spread is small, but a round trip pays it twice and gives up the final minute. For XSP, buying at the mid, other strikes, a higher fee and a cap on the premium all left the conclusion unchanged. SPY traded at the mid on both sides would have been profitable, but that is a price an order cannot count on (Section 7.2).

A vertical debit spread does not need a separate test. Bought at the long leg's ask and sold at the short leg's bid, it earns exactly the long leg's P&L at the ask, less the short leg's P&L at the ask, less the short leg's quoted spread and a fee, and every term but the spread is already in Table 9. With average spreads, no vertical did better per dollar of premium than its long leg alone.

11 An independent re-validation

After the studies were complete, the whole chain was rebuilt with new code that shares nothing with the study scripts and reads the same raw files. No new data were bought. The check is descriptive, because every day it uses had already been used. It can verify the earlier work and measure how much the data can say, but it cannot test a new setting. The re-validation reproduced the results, placed the start of the effect near a 60% body, and showed that the CFD candle from which the signals were computed differs in shape from the SPX index candle.

The first check was the integrity of the data. The index bars contain no duplicate, out-of-order or internally inconsistent bar, and every kept day has option quotes, or a record that no same-day expiry existed, and an official close. The weekdays without an official close are exactly the exchange holidays, the official closes equal the broker's daily closes on every day that has both, and every quote snapshot is stamped exactly on the minute, with no duplicate contract. At 15:51 the CFD price lies within about half an index point, on average, of the spot implied by put–call parity from the real quotes. The 15:50 bar is also the right minute: its body moves almost one for one with the option-implied change over the same minute, and not with the minutes before or after. One weakness was found. The CFD feed

was coarser in its early years, when between a fifth and a third of 15:50 bars had a body equal to the whole range, so a 60% threshold was a looser filter in the early discovery data than it is now.

Rebuilt from the raw quotes, the confirmation of Section 6 is identical trade for trade, and so are the 2013–2020 trades, the strike ladder, the yearly means, the late entries, the fixed-minute exits and the threshold profile of the re-tuning study. Coverage differs sharply between the two periods. Before 2021 the implied spot, which needs three strikes quoted on both sides, was often unavailable on calm days, so only a little over half of the signal days could be priced, against all but one since 2021. The missing days do not hide a better result: priced with the CFD price as the spot, as a sensitivity, they lost money at the ask as well (−$89 a trade, against +$68 at the mid). A five-day block bootstrap of 2021–2026 puts the current rule's mean between +$21 and +$115 a trade (90% interval), so the mean is positive but imprecisely measured.

Table 18: The 15:50 candle by body-to-range bucket.

	Index, 2013–2020		Index, 2021–2026		Options, 2021–2026		
Body / range	bps	t	bps	t	Trades	Ask	Mid
0.0–0.2	−1.46	−0.76	−0.29	−0.29	220	−53	−13
0.2–0.4	+0.20	+0.23	−0.22	−0.26	223	−21	+18
0.4–0.5	+1.38	+1.34	−0.16	−0.15	141	−15	+21
0.5–0.6	−2.35	−1.22	−0.24	−0.20	149	−59	−17
0.6–0.7	+3.97	+2.25	+1.71	+1.49	150	+6	+46
0.7–0.8	+2.77	+2.75	+1.26	+1.30	135	+42	+81
0.8–0.9	+0.73	+0.70	+3.50	+2.40	149	+97	+138
0.9–1.0	+2.17	+2.48	+2.92	+2.61	151	+122	+156
Below 0.6	−0.54	−0.71	−0.23	−0.46	733	−37	+2
At least 0.6	+2.42	+3.61	+2.36	+4.25	585	+67	+106

Note: Index move: CFD, 15:51 open to 15:59 close, in the candle's direction, in basis points, with a Newey–West (lag 5) t-statistic on that bucket's days. Option columns: ITM10 in the candle's direction, 2021–2026, dollars per trade at the 15:51 ask and at the mid. One trade lies exactly on the 0.80 boundary; classified in exact rather than floating-point arithmetic, it moves the option means at the ask of the 0.7–0.8 and 0.8–0.9 buckets to +$45 and +$94. Descriptive.

To see where the effect begins, Table 18 splits the 15:50 candle by the size of its body. Below 60% there is no effect in either period, in the index or in the option trade. From 60% up, the index moved in the candle's direction by about the same amount in both periods. Within the signal, larger bodies earned more per trade in 2021–2026, but no single option bucket is significant on its own, and the ordering is not stable: the 60–70% bucket, about break-even at the ask in 2021–2026, was the strongest in 2013–2020.

Above 60%, the thresholds cannot be ranked. Raising the threshold trades frequency for size, as Table 19 shows. Per day, the difference between 60% and 70% is statistically unresolved: no threshold differs significantly from 60% on the paired daily series, and the bootstrap interval for the difference between 0.70 and 0.60 straddles zero. The bootstrap shares in the table describe how unstable the in-sample optimum is; they are not probabilities that a given threshold is the best. A walk-forward choice, re-picking each year the threshold that had been best on all earlier days, earned less than a fixed 0.60 whichever history it

Table 19: The body threshold at real asks, 2021–2026.

Body at least	Trades	$/trade	$/day	t	Best
0.50	734	+42	+23.3	1.70	2%
0.55	656	+47	+23.2	1.80	1%
0.60 (current)	585	+67	+29.9	2.40	32%
0.65	508	+68	+26.3	2.27	2%
0.70	435	+89	+29.2	2.65	22%
0.75	380	+92	+26.4	2.56	7%
0.80	300	+110	+24.9	2.58	8%
0.85	219	+99	+16.5	1.94	1%
0.90	151	+122	+13.9	2.09	1%

Note: 15:50 candle, with the candle, ITM10 at the ask, 2021-01-01 to 2026-10-20. "Per day" means per session with a same-day expiry and prices (1,321 of 1,446 sessions). "Best" is the share of 5,000 five-day block-bootstrap resamples in which that threshold had the highest mean per day (grid 0.00 to 0.95 in steps of 0.05); the middle 90% of the resampled optimum ran from 0.25 to 0.80, and the 90% interval for 0.70 minus 0.60 is −$10.7 to +$8.8 a day. A walk-forward choice made +$16 a day on 2022–2026 when its history starts in 2021 and +$20 when it starts in 2013, against +$24 for a fixed 0.60 ($t = -0.98$ and −0.96 for the differences). 0.80 made +$38 a day on 2021–2023 and +$13 on 2024–2026. Descriptive.

started from, and with the longer history it chose 0.60 itself in most years. The two halves of 2021–2026 also disagree above 0.75. None of this is a fresh test, since the same days chose the rule. Within this sample the effect starts near 60%, and the thresholds from 60% to about 75% cannot be ranked.

The body-to-range ratio works as a cut-off and does not measure the strength of the signal. The bucket tables show where the effect sits, and correlations show how much of each day it explains (Table 20). Across all days a higher body-to-range ratio went with a larger index move in the candle's direction and a better option result, significantly but faintly: the largest of these correlations accounts for 0.7% of the day-to-day variation. Among signal days alone the relation is not distinguishable from zero in either period, and the thirds of the signal days by ratio show no stable slope (Table 21): in 2013–2020 the lowest third had the largest index move, and in 2021–2026 it had the smallest option result. Once a candle passes the threshold, a larger ratio does not mean a larger gain, which fits the finding that the thresholds above 60% cannot be ranked.

I looked at the candle's size, its body as a fraction of the index, only after the fact, and it is in no plan. On signal days a larger candle was followed by a larger move in its direction, in both periods and in the option result, but not by the right direction more often, and its rank correlations are smaller than its linear ones. A common volatility factor would produce this pattern, with larger candles in more volatile minutes followed by larger gains and larger losses, and the relation carried by a small number of large days. I treat it as a hypothesis for the forward record.

The re-validation also found a difference between the signal feed and the index. Every signal in this paper is read from the CFD candle, but a trader, and the live alert, reads the SPX index candle. The two track each other closely in level and direction, but a candle's body-to-range ratio depends on its shape, and the shapes differ. Real SPX one-minute bars are available for 36 sessions in the autumn of 2026, and the CFD series was extended over the same days. On those days the index candle showed a strong body at 15:50 more often

Table 20: Correlations between the 15:50 CFD candle and what followed.

Feature	Outcome	Days	Period	n	r	p	ρ	p
Body / range	Index move	All	2013–2020	1,785	+0.06	0.007	+0.04	0.066
Body / range	Index move	All	2021–2026	1,445	+0.08	0.002	+0.07	0.011
Body / range	Option result	All	2021–2026	1,318	+0.08	0.005	+0.07	0.018
Body / range	Index move	Signal	2013–2020	908	−0.05	0.108	−0.04	0.190
Body / range	Index move	Signal	2021–2026	644	+0.04	0.287	+0.02	0.605
Body / range	Option result	Signal	2021–2026	585	+0.06	0.177	+0.05	0.193
Body size	Index move	Signal	2013–2020	908	+0.19	< 0.001	+0.08	0.023
Body size	Index move	Signal	2021–2026	644	+0.13	0.001	+0.11	0.006
Body size	Right direction	Signal	2013–2020	901	+0.02	0.617	+0.03	0.438
Body size	Right direction	Signal	2021–2026	644	+0.04	0.255	+0.04	0.367
Body size	Option result	Signal	2021–2026	585	+0.12	0.003	+0.08	0.055

Note: Body size is $|c - o|/o$. Index move: CFD, 15:51 open to 15:59 close, in the candle's direction. Option result: ITM10 in the candle's direction at the 15:51 ask, held to settlement. "Right direction" is 1 if the index moved the candle's way. Signal days have a body of at least 60% of the range; days with a flat 15:50 candle are excluded. Periods: 2013-04-01 to 2020-12-31 and 2021-01-01 to 2026-10-20. r is Pearson's and ρ Spearman's coefficient; p-values assume independent days. Descriptive; body size was examined after the fact.

Table 21: Signal days in thirds by body-to-range ratio.

	2013–2020			2021–2026				
	Ratio	Days	bps	Ratio	Days	bps	Trades	$/trade
Lowest third	0.60–0.76	303	+3.95	0.60–0.74	215	+1.84	191	+19
Middle third	0.76–0.93	302	+1.38	0.74–0.87	214	+2.81	201	+102
Highest third	0.93–1.00	303	+1.91	0.87–1.00	215	+2.44	193	+80

Note: Thirds are formed within each period. Index move: CFD, 15:51 open to 15:59 close, in the candle's direction, in basis points. Option result: dollars per trade, ITM10 at the 15:51 ask, held to settlement. Descriptive.

than the CFD candle (Table 22). Every CFD signal was also an index signal, in the same direction, so no backtested signal was missed, but the index fired on six further days that the backtests would not have traded. Across all one-minute candles from 15:30 to 15:59 the pattern is the same: an index candle in the 60–70% range is a CFD strong body less than half the time, and only above 80% do the two nearly always agree.

A 60% threshold applied to the index candle is therefore a looser rule than the one that was tested; on this sample the tested rule fires about as often as a threshold of roughly 70% on the index candle. That correspondence concerns frequency only, and the sample is a single autumn. Whether the additional days pay was tested separately over 2021–2026 (Section 12): they did not, and even above 70% the two candles are not interchangeable.

The two ratios are also strongly correlated, at 0.90 for the 15:50 candle over the 36 sessions and 0.76 across all 1,080 candles from 15:30 to 15:59. The least-squares line through those candles, CFD ratio $= 0.10 + 0.72\times$ index ratio, puts the CFD's 0.60 at about 0.69 on the index. This is a second route to the correspondence above, and it does not change it.

The comparison continues in a two-feed forward record. Since 9 November 2026 the live alert records, every session, the 15:50 candle from both sources, each one's ratio and band

Table 22: The SPX index candle against the CFD candle, autumn 2026.

Index ratio	Candles	CFD ≥ 0.60	Cut-off	Days fired	CFD days	Others
0.0–0.4	398	5%	0.60	20	14	6
0.4–0.6	192	16%	0.65	18	14	4
0.6–0.7	143	42%	0.70	17	14	3
0.7–0.8	113	61%	0.75	16	14	2
0.8–0.9	98	82%	0.80	16	14	2
0.9–1.0	136	87%				

Note: 36 sessions, 2026-09-21 to 2026-11-09. Left: 1,080 one-minute candles from 15:30 to 15:59, by the index candle's body-to-range ratio, with the share that were CFD strong bodies. Right: the 15:50 candle, by the cut-off applied to the index candle; the CFD candle had 14 strong-body days, all among the index candle's. The median 15:50 ratio was 0.65 on the index candle and 0.54 on the CFD.

(below 0.60, 0.60–0.70, 0.70 or more) and the index move from the 15:51 open to the 15:59 close, and it labels each alerted signal with the index candle's band. The record starts with the 36 sessions already on file, of which only the 14 after 20 October lie outside every backtest (Table 23). It shows no difference yet: the index moved the candle's way about as often on sessions without a signal as on sessions with one, and on the six sessions where only the index candle reached 60% it did so four times. A correlation of the size found in the history needs on the order of a thousand sessions to be detected, and the bands need a few hundred signals before they can be compared. Section 12 adds a more specific question for the record, whether sessions on which both candles show a strong body continue to outperform sessions on which only the index candle does. Since the study of Section 13, the record also holds the SPY candle, and the alert acts on the rule described at the end of that section.

Table 23: The two-feed record, autumn 2026.

	Index candle			CFD candle		
Body / range	Days	Right	Move	Days	Right	Move
0.70 or more	17	11	+1.53	13	8	+1.55
0.60–0.70	3	2	+3.87	1	1	+1.24
Below 0.60	16	10	+1.08	22	14	+1.53

Note: 36 sessions, 2026-09-21 to 2026-11-09, by the band of the 15:50 candle on each source; only the 14 sessions after 2026-10-20 lie outside every backtest. "Right" counts sessions on which the index moved in the candle's direction from the 15:51 open to the 15:59 close; "Move" is the mean of that move, in index points. Descriptive; too few sessions to compare bands.

12 The rule on the index candle

The re-validation left one question open. Every signal in this paper was read from the CFD candle, while a trader, and the live alert, reads the SPX index candle, and Section 11 showed that the two differ in shape. A further pre-registered study therefore read the 15:50 candle from an archive of one-minute SPX index bars published by CNBC, the source of the live alert, and traded it at the option prices of Section 6. The plan was locked before any index candle had been joined with a price. The option outcomes of these days were already known, so the study is, in the plan's words, an outcome-informed, retrospective check of the signal's source: not an independent confirmation and not a new holdout. Read from the index candle,

the rule did not meet its criterion, and the gain was confined to the days on which both candles showed a strong body.

The archive holds a valid 15:50 bar on 1,310 of the 1,322 sessions from 2021-01-01 to 2026-10-20 that have option quotes and an official close (99.1%). Before the lock, and using price levels only, the archived bar was compared with the broker's own index bar on the 22 sessions of the period that have both, and all four prices were equal on 21. In every year the archived 15:50 close lay nearer the CFD's 15:50 close than the CFD's closes a minute earlier or later, which confirms the time stamps. The signal is the live alert's test, computed in whole cents: a body of at least 60% of the range, split into a 60–70% band and a band of 70% or more. The trade is unchanged: ITM10 in the candle's direction at the 15:51:00 ask, held to settlement. The primary line asks whether the rule read from the index candle has a positive mean per day with Newey–West $t \geq 2.0$. Three secondary lines, one for each band and one for the difference between them per trade by Welch's test, also require a Holm-adjusted $p \leq 0.05$.

Table 24: The rule read from the SPX index candle, 2021–2026.

Index candle	Trades	$/trade	At mid	Won	$/day	t	Holm p	Reading
Body ≥ 60%	710	+$39.44	+$78.14	52%	+$21.39	+1.53		criterion not met
Body ≥ 70%	579	+$53.73	+$91.20	53%	+$23.75	+1.86	0.094	insufficient evidence
Body 60–70%	131	−$23.69	+$20.44	49%	−$2.37	−0.41	0.685	insufficient evidence

Note: One ITM10 contract at the 15:51 ask, held to settlement, 2021-01-01 to 2026-10-20. The first row is the primary line. Per day is the mean of the daily series (0 on days outside the cell); t is Newey–West, lag 5. Holm p is adjusted across the three secondary lines, the third of which, the difference between the bands, is given in the text. Pre-registered.

Table 24 gives the registered lines. Read from the index candle, the rule made +$39.44 a trade, against +$67.24 from the CFD candle on the same days (583 trades), and its daily series fell short of the criterion ($t = 1.53$). The source-check criterion was therefore not met. The band of 70% or more came closer and also fell short. The 60–70% band lost a little at the ask and made a little at the mid; under the plan this is insufficient evidence in either direction, and the difference between the two bands (+$77.41 a trade, Welch $t = 1.20$) cannot be distinguished from zero. By year (Table 28) the 60–70% band alternated between large losses and large gains with no pattern, the band of 70% or more lost money only in 2023, and the rule as a whole made almost nothing from 2022 to 2024.

Table 25: Where the two candles disagree, 2021–2026.

						Index ≥ 70%		Index 60–70%	
Strong body on	Trades	Ask	Mid	Won	t	Trades	Ask	Trades	Ask
Both candles	520	+$80.95	+$120.14	54%	+2.51	481	+$86	39	+$14
Index candle only	190	−$74.14	−$36.80	45%	−1.68	98	−$107	92	−$40
CFD candle only	63	−$45.92	−$15.20	51%	−0.73	0		0	

Note: ITM10 in the index candle's direction (in the CFD candle's direction on days only the CFD candle signals), $ per trade at the ask and at the mid; t is a simple t-statistic. The two candles never signalled opposite directions. Descriptive.

The analysis of where the difference comes from is descriptive. The two candles never pointed in opposite directions, but they often disagreed about strength (Table 25), and the

gain was confined to the days on which both showed a strong body. The 190 trades taken on days when only the index candle did lost $74.14 a trade, and they include index candles of 70% or more. Reading the index candle therefore adds a set of days without the effect, and these lower the mean. The correspondence of Section 11 still holds as a statement about frequency: over 1,310 days the two ratios have a correlation of 0.78, and the CFD's 0.60 corresponds to 0.69 on the index candle. It does not make the candles interchangeable, since 17% of index candles of 70% or more were below 60% on the CFD. One explanation, which I did not test, is that the CFD follows the futures, which move first, while the index is computed from the last trades in 500 stocks and can show a strong candle while it catches up.

By the size of the index candle's body (Table 26), the option trade was profitable only from 80% upward (438 trades, +$89 a trade); that cut was not planned. The index itself moved in the candle's direction after candles of 70% or more and not after candles of 60–70% (Table 27), so the directional effect is present on the index candle, and only the option trade falls short.

Table 26: Option results by the index candle's body-to-range ratio, 2021–2026.

Body / range	Days	Trades	Ask	Mid	Won
0.0–0.1	80	79	+$78	+$118	53%
0.1–0.2	90	90	−$21	+$17	50%
0.2–0.3	93	92	−$23	+$17	46%
0.3–0.4	105	105	−$42	−$5	50%
0.4–0.5	115	115	−$53	−$17	44%
0.5–0.6	113	113	−$12	+$28	52%
0.6–0.7	132	131	−$24	+$20	49%
0.7–0.8	141	141	−$56	−$15	47%
0.8–0.9	160	160	+$67	+$104	54%
0.9–1.0	278	278	+$102	+$137	54%

Note: ITM10 in the candle's direction whatever the ratio, $ per trade. A ratio of exactly 1 is in the last row. Descriptive.

Table 27: The index move after the index candle, 2021–2026.

Index candle	Days	bps	t	Right
Body ≥ 70%	640	+2.63	+4.55	57%
Body 60–70%	141	+0.14	+0.11	51%
Body < 60%	650	+0.23	+0.46	53%

Note: 15:51 open to 15:59 close of the archived index bars, in the candle's direction, in basis points; t is Newey–West, lag 5; "Right" is the share of non-zero moves in the candle's direction. Descriptive.

Table 28: The rule read from the index candle, by year.

Signal on the index candle	2021	2022	2023	2024	2025	2026
Body ≥ 60%	+$105 (88)	+$7 (112)	+$5 (136)	+$7 (125)	+$51 (134)	+$83 (115)
Body ≥ 70%	+$218 (63)	+$45 (91)	−$19 (110)	+$32 (102)	+$19 (118)	+$104 (95)
Body 60–70%	−$179 (25)	−$157 (21)	+$109 (26)	−$104 (23)	+$285 (16)	−$16 (20)

Note: $ per trade at the ask, with the number of trades in parentheses. Descriptive.

The study has limits. The days and their option outcomes are those of the earlier confirmation; only the candle changed, so the registered lines test the source of the signal and nothing else. The archive is a vendor's stored bar, which need not equal the bar shown live, and 12 sessions had none; on those the CFD rule's known result was ordinary (+$22 a day against +$30 on the others). The disagreement table, the tenths and the 80% cut were read after the result. As in every real-price study, entry is at the quote of the instant the candle closes.

13 The rule on the SPY candle

The index-candle study left a practical problem. The candle that met its criterion, the CFD's, cannot be seen live, and the candle that can, the index's, did not meet it. The CFD follows the futures, and SPY, the exchange-traded fund, trades live and follows them as well. A further pre-registered study therefore read the 15:50 candle from archived one-minute SPY bars and traded the same SPXW option at the same prices. SPY was the third candle source evaluated against the same option outcomes, and the study's second line was chosen after the agreement of two candles had been seen to do well. The plan was locked before any SPY candle was joined with a price, which prevents further adaptation but does not remove that selection. Its lines are therefore retrospective numerical criteria, not an independent confirmation, a new holdout or an established trading edge. Read from the SPY candle, the rule met those criteria, and the gain was again confined to sessions on which all three candles showed a strong body.

The archive, from CNBC as for the index, holds a valid SPY 15:50 bar on 1,312 of the 1,322 sessions with option quotes and an official close (99.2%). Checks made before the lock, on price levels only, showed that the bar labelled 15:50 is the same minute as the index bar in every year, that the bars are traded prices and not dividend-adjusted (their 15:59 close equals the unadjusted daily close at the median in every year), and that 16% of the prices are sub-penny prints, which are rounded to the cent. They also showed that an ETF candle depends on its vendor: against a second vendor's SPY bars on 22 sessions, the four prices of the 15:50 bar agreed to the cent on 14, with differences of up to $0.02 elsewhere. The signal is the same whole-cent test applied to the SPY bar, and the trade is ITM10 in the SPY candle's direction at the 15:51:00 ask, held to settlement. The primary line asks whether the daily series has a positive mean with Newey–West $t \geq 2.0$. Two secondary lines, the SPY and index candles both signalling in the same direction and the SPY candle's band of 70% or more, also require a Holm-adjusted $p \leq 0.05$.

Table 29 shows that all three registered lines met their retrospective criteria. Read from the SPY candle, the rule made +$71.74 a trade ($t = 2.35$), about what the CFD candle made on the same days and nearly twice what the index candle made. The band of 70% or more was the strongest line and was positive in each of the six years (Table 30); the 60–70% band made nothing, as on the other two sources.

The comparison of the three candles is descriptive. A strong SPY candle is nearly always a strong CFD candle: 94% of SPY candles of 70% or more were CFD signals (Table 31). The gain was confined to sessions on which all three candles showed a strong body (435 trades, +$112.78 a trade; Table 32), and every other combination lost on average, including the index and CFD candles signalling without SPY. No single row of that table is significant

Table 29: The rule read from the SPY candle, 2021–2026.

Signal	Trades	$/trade	At mid	Won	$/day	t	Holm p	Reading
SPY body $\geq$ 60%	523	+$71.74	+$108.51	53%	+$28.62	+2.35		criterion met
SPY and index both $\geq$ 60%	504	+$80.44	+$117.34	54%	+$31.21	+2.56	0.0053	criterion met
SPY body $\geq$ 70%	369	+$103.40	+$137.67	55%	+$29.08	+2.84	0.0045	criterion met
SPY body 60–70%	154	−$4.14	+$38.66	49%				descriptive
CFD rule, same days	582	+$68.84	+$107.33	54%				descriptive

Note: One ITM10 contract at the 15:51 ask in the SPY candle's direction, held to settlement, 2021-01-01 to 2026-10-20. The first row is the primary line and the next two are the secondary lines; Holm p is adjusted across those two. Per day is the mean of the daily series; t is Newey–West, lag 5. The last two rows are descriptive. Pre-registered, retrospective.

Table 30: The rule read from the SPY candle, by year.

Signal	2021	2022	2023	2024	2025	2026
SPY body $\geq$ 60%	+$201 (51)	+$46 (78)	+$47 (98)	+$60 (98)	+$44 (107)	+$93 (91)
SPY and index both $\geq$ 60%	+$210 (49)	+$81 (75)	+$50 (92)	+$71 (95)	+$44 (103)	+$92 (90)
SPY body $\geq$ 70%	+$281 (34)	+$141 (46)	+$30 (70)	+$94 (76)	+$95 (79)	+$84 (64)
SPY body 60–70%	+$40 (17)	−$91 (32)	+$91 (28)	−$56 (22)	−$102 (28)	+$117 (27)

Note: $ per trade at the ask, with the number of trades in parentheses. Descriptive.

on its own, but together they repeat the finding of Section 12 that a strong body on only some of the candles carries no edge. By the size of the SPY candle's body (Table 33), the trade was profitable in every tenth from 70% upward, a lower cut than the index candle's 80%; below 70% only one tenth, 0.1–0.2, was positive. The index moved in the SPY candle's direction after SPY candles of 70% or more (Table 34).

Table 31: Where a SPY candle's band falls on the other two candles.

Other candle	SPY band	Days	Other $<$ 60%	Other 60–70%	Other $\geq$ 70%
CFD	SPY $\geq$ 70%	369	6%	11%	83%
CFD	SPY 60–70%	155	35%	33%	32%
CFD	SPY $<$ 60%	788	83%	7%	10%
Index	SPY $\geq$ 70%	366	0%	2%	98%
Index	SPY 60–70%	154	9%	18%	73%
Index	SPY $<$ 60%	780	74%	12%	14%

Note: The share of each SPY band's sessions in each band of the CFD candle and of the index candle, 2021–2026. Descriptive.

Several limits apply. SPY is the third source tried on the same outcomes, so its t-statistics overstate what a fresh sample would support, and the days are not new. The result belongs to one vendor's archived SPY candle: a second vendor's candle differed by a cent or two on about a third of the sessions compared, enough to move a borderline candle across a band edge. The 10 sessions without an archived SPY bar were poor days for the CFD rule (−$60 a day against +$31 on the rest), so the estimates are slightly flattered. The three-candle table, the tenths and the yearly figures were read after the result, and entry is at the quote of the instant the candle closes. Since the run, the live alert reads the SPY candle as well as the index candle and records both, so the comparison continues on new days.

The rule now in use is descriptive and was not registered. After the SPY study the live

Table 32: Which of the three candles show a strong body, 2021–2026.

SPY	Index	CFD	Days	Trades	Ask	Mid	Won	t
yes	yes	yes	436	435	+$112.78	+$150.18	56%	+3.11
yes	yes	–	69	69	−$123.42	−$89.66	36%	−1.73
yes	–	yes	9	9	−$118.42	−$91.20	33%	−0.80
yes	–	–	6	6	−$401.42	−$378.09	33%	−1.25
–	yes	yes	82	82	−$77.58	−$28.55	44%	−1.20
–	yes	–	121	121	−$46.04	−$6.66	50%	−0.82
–	–	yes	54	54	−$33.83	−$2.53	54%	−0.49
–	–	–	523	0				

Note: ITM10 in the direction of the first signalling candle in the order SPY, index, CFD; 1,300 sessions with all three candles; $ per trade at the ask and at the mid; t is a simple t-statistic. No session had signalling candles in opposite directions. Descriptive.

Table 33: Option results by the SPY candle's body-to-range ratio, 2021–2026.

Body / range	Days	Trades	Ask	Mid	Won
0.0–0.1	91	91	−$51	−$5	53%
0.1–0.2	110	109	+$119	+$161	61%
0.2–0.3	124	124	−$28	+$10	45%
0.3–0.4	144	143	−$87	−$49	45%
0.4–0.5	137	137	−$37	+$5	53%
0.5–0.6	161	161	−$74	−$35	45%
0.6–0.7	155	154	−$4	+$39	49%
0.7–0.8	145	145	+$90	+$124	54%
0.8–0.9	125	125	+$137	+$174	56%
0.9–1.0	99	99	+$81	+$112	55%

Note: ITM10 in the candle's direction whatever the ratio, $ per trade. A ratio of exactly 1 is in the last row. Descriptive.

Table 34: The index move after the SPY candle, 2021–2026.

SPY candle	Days	bps	t	Right
SPY body ≥ 70%	399	+3.42	+4.70	58%
SPY body 60–70%	166	+1.26	+1.20	56%
SPY body < 60%	835	+0.35	+0.74	55%

Note: 15:51 open to 15:59 close of the archived index bars, in the SPY candle's direction, in basis points; t is Newey–West, lag 5; "Right" is the share of non-zero moves in the candle's direction. Descriptive.

alert was changed to act on one rule built from its lines: the SPY 15:50 candle in its band of 70% or more, and the index candle at 60% or more in the same direction; ITM10 in that direction at the 15:51 ask, held to settlement. This is the intersection of two cells the study reported, not a registered line of its own, and it is nearly the 70% band itself: of that band's 369 trades, 3 had no index bar and 1 had an index candle below 60%, which leaves 365. On 2021-01-01 to 2026-10-20 those trades made +$101.33 each and +$36,986 in all, with 54.5% profitable, an average gain of $593 against an average loss of $488, a worst trade of −$2,210, a longest run of 7 losing trades and a worst peak-to-trough decline of $5,271 (Table 35). The rule traded on about 30% of sessions once expiries became daily. These figures describe days that helped to shape the rule, with the entry at the instant the candle closes, and they are

not evenly spread: from 2026-08-10 to 2026-10-20, the last ten weeks of the sample, the rule lost $974 over 18 trades, 7 of them profitable. The 7 signal days after the sample (2026-10-22 to 2026-11-06) were too few to mean anything: 4 priced from the collector's one-minute broker bars made +$1,214 and 3 priced from OPRA asks bought afterwards made +$562, +$1,775 in all with 5 of 7 profitable. The alert now sends two messages a session, whether the candles look like arming at about 15:50:40 and the decision right after 15:51:00, and logs every session's three candles and the outcome.

Table 35: The rule now in use, by year.

Year	Sessions	Trades	Share	$/trade	Total	Won
2021	251	33	13%	+$291	+$9,595	20
2022	250	45	18%	+$140	+$6,309	28
2023	248	68	27%	+$15	+$1,050	36
2024	249	76	31%	+$94	+$7,150	40
2025	247	79	32%	+$95	+$7,527	45
2026	201	64	32%	+$84	+$5,355	30
All	1,446	365	25%	+$101	+$36,986	199

Note: SPY 15:50 candle of 70% or more and index candle of 60% or more in the same direction, ITM10 at the 15:51 ask, held to settlement, one contract. Sessions are full sessions; expiries were not daily before 2022. Won is the number of profitable trades. Descriptive; not a registered line.

14 A search over closing candles

The studies so far fixed the candle and its cut and asked whether that rule worked. A last pre-registered study asked a broader question, namely which setting among the closing candles, cuts, directions and strikes near the rule does best on real asks, and whether the best one is better than luck. The plan was locked after every earlier study, and two of its three candle sources, the SPY candle and the agreement of the SPY and index candles, were chosen after their results had been seen. It therefore guards against further selection but not against that history, and it set aside in advance every 15:50 setting, because those had already been examined on the later years and could not be tested on them afresh. The search chose a 15:50 setting again, so nothing went to the test, and no other closing minute beat the 15:50 candle on the tuning years.

The search covers 7,128 settings: three candle sources (the index candle, the SPY candle, and the two agreeing in direction), the eleven one-minute candles from 15:45 to 15:55, twelve body cuts from 40% to 95%, trading with or against the candle, and nine strikes from ITM20 to OTM20. Each trade buys one SPXW contract at the ask in the snapshot that opens the next minute and holds it to settlement. A setting is scored on 2021–2023 by its mean profit per day averaged with the settings one cut above and below, so that an isolated spike does not win. To be eligible it needs at least 100 trades, at least 40 in each half of the period, a positive mean in each half, and prices for at least 95% of its signals; 387 settings qualified. The best one would be tested once on 2024–2026 only if it was not a 15:50 setting, if a luck diagnostic gave $p \leq 0.05$, and if it beat the index-candle rule of Section 12 on the tuning days. The luck diagnostic repeats the whole search 195 times with every candle signal moved by a whole number of sessions against fixed outcomes, which keeps each signal's own pattern

but breaks its link to the outcome. Before any statistic, the code reproduced the trades of the two candle studies exactly (336 and 227 on the tuning days, 374 and 296 on the test days). It also passed synthetic self-tests and a full trial on synthetic data.

Table 36: The best settings on 2021–2023.

Source	Candle	Body	Side	Strike	Score	$/day	Trades	First half	Second half
Index	15:50	$\geq 80\%$	With	ITM20	+$37.46	+$39.42	193	+$77.15	+$1.08
Index	15:50	$\geq 80\%$	With	ITM15	+$33.87	+$35.50	192	+$69.96	+$0.59
Index	15:54	$\geq 60\%$	With	ITM5	+$33.55	+$32.43	299	+$50.12	+$14.33
Both	15:54	$\geq 40\%$	With	ITM5	+$32.88	+$38.17	325	+$53.11	+$23.03
Index	15:54	$\geq 65\%$	With	ITM5	+$32.47	+$32.71	261	+$50.36	+$14.71
Both	15:50	$\geq 65\%$	With	ITM20	+$32.19	+$34.72	179	+$67.80	+$1.09
Index	15:54	$\geq 70\%$	With	ITM5	+$31.94	+$32.26	224	+$45.21	+$18.97
Index	15:50	$\geq 80\%$	With	ITM5	+$31.88	+$33.85	192	+$67.00	+$0.27
SPY	15:54	$\geq 40\%$	With	ITM5	+$31.46	+$35.46	342	+$49.97	+$20.76
Both	15:54	$\geq 45\%$	With	ITM5	+$31.40	+$27.58	283	+$45.98	+$8.75

Note: The ten eligible settings with the highest score, of 387 eligible among 7,128. Score is the mean profit per day averaged with the same setting one cut above and below; $/day is the setting's own mean over 625 sessions. The halves are 2021-01-01 to 2022-09-30 and 2022-10-01 to 2023-12-31. One contract at the ask, held to settlement, $0.09 in fees. Pre-registered.

Table 36 shows that the search chose a 15:50 setting: the index candle with a body of 80% or more, traded in its direction with ITM20. Because 15:50 settings were set aside, nothing went to the test, and this choice can be checked only on later sessions. It also missed the luck bar: the shifted searches reached its score in 9 of 195 cases ($p = 0.051$; median best score +$20.11, 95th percentile +$37.38). Its gain came almost entirely in the first half of the tuning period (+$77.15 a day against +$1.08). By candle minute (Table 37), 15:50 led and 15:54 came second, every other minute scored below both, and two minutes had no eligible setting at all.

Table 37: The best eligible setting for each candle, 2021–2023.

Candle	Source	Body	Side	Strike	Score	$/day	Trades
15:45	Index	$\geq 80\%$	With	ITM20	+$21.46	+$27.65	166
15:46	Index	$\geq 40\%$	With	ITM5	+$1.48	+$6.81	430
15:47	Index	$\geq 70\%$	With	ITM5	−$0.19	+$4.25	235
15:48	Index	$\geq 40\%$	With	ATM	+$19.30	+$19.41	416
15:49	Index	$\geq 90\%$	With	ATM	+$5.22	+$7.75	104
15:50	Index	$\geq 80\%$	With	ITM20	+$37.46	+$39.42	193
15:51	No eligible setting						
15:52	Both	$\geq 50\%$	Against	ITM5	+$12.00	+$13.31	237
15:53	Index	$\geq 80\%$	With	ATM	+$23.40	+$23.69	177
15:54	Index	$\geq 60\%$	With	ITM5	+$33.55	+$32.43	299
15:55	No eligible setting						

Note: As in Table 36, the highest-scoring eligible setting among those that read the given candle. Entry is at the next minute. Pre-registered.

The ranking over the whole period is descriptive. After the decision, the grid was ranked on all of 2021–2026 by mean profit per day, among settings with at least 200 trades and prices for 95% of their signals. All fifteen leading settings were 15:50 candles traded in their

direction, 10 of them with ITM20 (Table 38). The first, Both 15:50, $\geq$ 60%, with, ITM20, made +$34.74 a day (+$89.44 a trade, $t = 2.71$, 505 trades), above the 95th percentile of the same ranking repeated on 427 shifted signals (+$30.67). The ranking reuses days already seen and is not a test. Table 39 follows the 15:50 settings examined in earlier sections across the two periods. Every one made money in both, the 60% rule least on the tuning days, and on 2024–2026 only the SPY candle at 70% with ITM20 reached $t = 2$. Paying the ask was costly throughout: at the mid, the search's choice made +$61.03 a day on 2021–2023 and +$41.32 on 2024–2026, against +$39.42 and +$29.58 at the ask.

Table 38: The leading settings over 2021–2026, all on the 15:50 candle.

Source	Body	Strike	$/day	t	Trades	$/trade	Won
Both	$\geq$ 60%	ITM20	+$34.74	+2.71	505	+$89.44	55%
Index	$\geq$ 85%	ITM20	+$34.23	+3.23	365	+$122.87	55%
Index	$\geq$ 80%	ITM20	+$34.20	+2.92	438	+$102.29	55%
Both	$\geq$ 65%	ITM20	+$33.97	+2.85	429	+$102.93	55%
SPY	$\geq$ 65%	ITM20	+$33.44	+2.83	446	+$98.37	55%
SPY	$\geq$ 70%	ITM20	+$33.08	+3.11	369	+$117.61	55%
Index	$\geq$ 75%	ITM20	+$32.23	+2.67	500	+$84.44	54%
Both	$\geq$ 70%	ITM20	+$32.20	+3.01	357	+$117.25	55%
Both	$\geq$ 60%	ITM15	+$32.10	+2.53	503	+$82.84	55%
SPY	$\geq$ 60%	ITM20	+$31.92	+2.49	524	+$79.92	54%

Note: The ten settings with the highest mean profit per day on 2021-01-01 to 2026-10-20 among 3,887 with at least 200 trades and prices for 95% of their signals; all trade in the candle's direction. t is Newey–West, lag 5. Ranked after the tuning decision on days already seen. Descriptive.

Table 39: The 15:50 settings examined earlier, by period.

	2021–2023			2024–2026		
Setting	$/day	t	Trades	$/day	t	Trades
Index $\geq$ 60%, ITM10 (Section 12)	+$17.54	+0.87	336	+$24.80	+1.27	374
Index $\geq$ 80%, ITM10	+$37.26	+2.49	193	+$23.05	+1.39	245
Index $\geq$ 80%, ITM20	+$39.42	+2.50	193	+$29.58	+1.73	245
Index $\geq$ 85%, ITM10	+$35.50	+2.73	161	+$22.85	+1.47	204
Index $\geq$ 85%, ITM20	+$38.29	+2.83	161	+$30.64	+1.92	204
SPY $\geq$ 70%, ITM10	+$29.28	+2.29	150	+$28.91	+1.85	219
SPY $\geq$ 70%, ITM20	+$30.56	+2.26	150	+$35.32	+2.20	219
SPY $\geq$ 80%, ITM10	+$22.50	+2.12	85	+$16.22	+1.22	139
SPY $\geq$ 80%, ITM20	+$23.31	+2.13	85	+$21.11	+1.53	139
Both $\geq$ 60%, ITM10	+$34.52	+2.11	216	+$28.30	+1.58	288
Both $\geq$ 60%, ITM20	+$32.54	+1.84	217	+$36.69	+1.99	288

Note: The 15:50 candle traded in its direction, one contract at the ask, held to settlement. The first row is the index-candle rule; the rest are the settings the plan set aside as examined. "Both" requires the index and SPY candles to pass the cut in the same direction. t is Newey–West, lag 5. Descriptive.

The question of which candle predicts the index move is also descriptive. Without options, the same signals can be scored on the index move itself, from the next minute's open to the 15:59 close in the candle's direction (Table 40). On 2021–2026 the ten largest t-statistics all belonged to 15:50 candles, led by the index candle at 85% (+3.52 basis points over 408

sessions, $t = 5.21$). On 2019–2020, with the index candle alone, no 15:50 setting was among the ten largest, and the largest t (2.90, on 23 sessions) was below the median of its shifted benchmark. The registered benchmark for this table, the 95th percentile of the largest t on shifted signals (5.57 on 2021–2026), was not cleared. A check chosen after the run shows why the bar is high: in 62% of the shifted tables the largest t came from a setting with fewer than 100 sessions, and among settings with at least 300 sessions the shifted 95th percentile was 3.40. The 15:50 result is therefore strong for its sample size, but it did not pass the bar fixed in advance.

Table 40: The index move after each closing candle.

Source	Candle	Body	Sessions	bps	t	Right
Panel A: 2019–2020, index candle only						
Index	15:47	$\geq 95\%$	23	+6.78	+2.90	57%
Index	15:53	$\geq 90\%$	31	+7.95	+2.85	61%
Index	15:48	$\geq 75\%$	60	+5.11	+2.78	60%
Index	15:48	$\geq 80\%$	49	+5.01	+2.65	59%
Index	15:48	$\geq 70\%$	73	+4.32	+2.57	59%
Panel B: 2021–2026, all three sources						
Index	15:50	$\geq 85\%$	408	+3.52	+5.21	58%
Index	15:50	$\geq 80\%$	490	+3.35	+5.21	59%
Index	15:50	$\geq 75\%$	555	+2.99	+5.01	58%
Both	15:50	$\geq 65\%$	468	+3.38	+4.94	59%
SPY	15:50	$\geq 65\%$	482	+3.26	+4.91	59%
Both	15:50	$\geq 60\%$	550	+2.97	+4.79	58%
SPY	15:50	$\geq 70\%$	399	+3.42	+4.70	58%
SPY	15:50	$\geq 60\%$	565	+2.78	+4.61	57%
Both	15:50	$\geq 70\%$	390	+3.43	+4.59	58%
Both	15:50	$\geq 55\%$	629	+2.61	+4.58	57%

Note: The settings with the largest t, of 132 on 2019–2020 and 396 on 2021–2026: the index move from the open of the minute after the candle to the 15:59 close, in the candle's direction, in basis points, on sessions on which the candle passed the cut. t is Newey–West, lag 5; "Right" is the share of non-zero moves in the candle's direction. Shifted benchmark for the largest t: median 3.26 and 95th percentile 6.39 on 2019–2020, 3.00 and 5.57 on 2021–2026. Descriptive.

The search also has limits. It reuses the 2021–2026 outcomes of the earlier studies, and two of its candle sources were chosen with knowledge of them, so its luck diagnostic corrects for the 7,128 settings but not for that history. Its choice was not clearly beyond luck and earned almost all of its tuning gain before October 2022. The search is still useful as a comparison. No other closing minute, cut, direction or strike beat the 15:50 candle on 2021–2023, and settings that ask for a cleaner candle (80% or more on the index, 70% or more on SPY, or the two candles agreeing) made more than the 60% rule on 2021–2023 and between +\$16.22 and +\$36.69 a day on 2024–2026, against its +\$24.80. Those are hypotheses that only sessions after November 2026 can test.

15 Closing-auction imbalances

From 15:50 ET the New York Stock Exchange publishes the closing-auction imbalance of the stocks it lists, and Nasdaq disseminates an early imbalance indicator for its own stocks

every ten seconds.[1] The 15:50 candle is therefore the first full minute in which the market can react to the published imbalance. That suggests an explanation of the signal: traders act on the publication during the 15:50 minute, and the index keeps drifting the same way while the auction absorbs the imbalance. If that channel mattered, the rule should be right more often on days when its candle points the same way as the published imbalance than on days when it points the other way. A pre-registered study (plan locked 2026-11-10 19:37 UTC, one registered run) tested this prediction on the rule's 644 signal days of 2021–2026, the days of its holdout.

For each year the study used the 50 largest common-stock positions of SPY at the previous year-end, from the fund's N-PORT reports to the SEC; they held between 51% and 63% of its net assets. On each signal day it bought from Databento the imbalance messages that each stock's listing exchange sent between 15:50 and 16:00 (NYSE Pillar for NYSE-listed stocks, Nasdaq TotalView-ITCH for Nasdaq-listed ones). For each stock it kept the first closing-auction imbalance record received in the 15:50 minute and summed the signed dollar imbalances, $D = \sum_i s_i Q_i p_i$, where s_i is $+1$ for a buy imbalance, -1 for a sell imbalance and 0 for none, Q_i is the imbalance in shares and p_i the reference price, in exact integer arithmetic. A day with at least 10 usable stocks and $D \neq 0$ is classified, and it *agrees* if the sign of D is the candle's direction. The outcome is the rule's own: a hit when the CFD moves from the 15:51 open to the 15:59 close in the candle's direction. The primary test (H1) compares the hit rates on agreeing and disagreeing days, $d = h_a - h_d$, and passes if $d > 0$ and $t = d/\mathrm{se} \geq 2.0$, with the standard error of two independent proportions. Two secondary tests, which decide nothing, ask whether the candle agrees with the imbalance on more than half of the days (H2) and whether the imbalance direction alone predicts the move (H3), with a Holm correction. A block bootstrap gives an interval for d, and the robustness checks of Table 41 were fixed in the plan, as was the wording of the verdict. By the plan's power calculation, which treats days as independent, 80% power at the bar needs a true d of about 11 to 13 points, so the test can detect only a large concentration of the edge on agreeing days.

Every signal day was classified. All 50 stocks had a usable first record on each of the 644 days, and the first closing record of the minute arrived a median of about 0.1 seconds after 15:50:00 on both exchanges (at most 1.1 seconds).

Table 41 reports the results. The candle and the published imbalance usually pointed the same way: they agreed on 469 of the 644 days (72.8%), far more often than half (H2, $p = 3.1 \times 10^{-32}$), but agreement did not make the candle more accurate. It was right on 55.9% of agreeing days and on 61.7% of disagreeing days, so $d = -5.9$ points (se 4.3, $t = -1.35$), and H1 did not pass. The block-bootstrap interval of d, from -14.8 to $+4.0$ points, lies below the 11 to 13 points that the test was designed to detect. The imbalance direction alone was right on 51.1% of days (H3, $p = 0.30$). The robustness checks agree: the difference in hit rates stayed negative with the latest record of the minute, with NYSE or Nasdaq stocks alone and within year and direction. The average move in the candle's direction was about the same on both kinds of day (2.41 against 2.24 basis points), and the option trade made

[1]The NYSE moved its market-on-close cut-off and imbalance publication from 15:45 to 15:50 on 1 April 2019 (SEC Release No. 34-85021). Nasdaq's early indicator from 15:50 was scheduled for the fourth quarter of 2019 (Release No. 34-87179). Samples of the exchanges' messages on 30 December 2020 and 6 November 2026, outside the sample, confirmed both schedules.

Table 41: The closing-imbalance test: the rule's hit rate on days that agree and disagree with the 15:50 imbalance.

	Agreeing days	Disagreeing days	Difference	Statistic
Panel A: registered tests				
H1: hit rate (primary)	55.9% (469)	61.7% (175)	−5.9	$t = -1.35$
Block-bootstrap 95% interval			−14.8 to +4.0	
H2: candle agrees with imbalance	469 of 644 (72.8%)			$p = 3.1 \times 10^{-32}$
H3: imbalance direction right	329 of 644 (51.1%)			$p = 0.30$
Panel B: robustness				
Latest record before 15:51	55.6% (459)	62.2% (185)	−6.6	$t = -1.55$
NYSE-listed stocks only	56.2% (372)	59.2% (272)	−3.0	$t = -0.76$
Nasdaq-listed stocks only	57.3% (499)	57.9% (145)	−0.6	$t = -0.13$
Adjusted for year and direction			−6.4	12 cells
Move in the candle's direction (bp)	2.41 (469)	2.24 (175)	+0.16	$t = 0.13$
Option P&L, $ a trade	+$80.43 (423)	+$33.58 (162)	+$46.85	$t = 0.79$

Note: Signal days 2021–2026; all 644 were classified. Cells give the value and, in brackets, the number of days (trades for the option P&L). Differences of hit rates are in percentage points; t uses the standard error of two independent proportions, and the move and option P&L rows use Welch's t. The bootstrap draws 20-day blocks of the 1,446 sessions, 10,000 of 10,000 draws kept. H2 and H3 are exact one-sided binomial tests against one half (the half benchmark is descriptive), with a Holm correction: H2 rejects, H3 does not reject. Panel B was fixed in the plan and cannot change the verdict. Move: from the 15:51 open to the 15:59 close in the candle's direction, basis points. Option P&L: ITM10 at the 15:51 ask, held to settlement, $0.09 fee, on the 585 priced trades.

Table 42: Signal days that agree and disagree with the 15:50 closing imbalance, by year.

		Agreeing days			Disagreeing days		
Year	Days	n	Hit rate (95% CI)	bp	n	Hit rate (95% CI)	bp
2021	111	92	62.0% (52–71)	4.67	19	78.9% (57–91)	5.12
2022	105	76	57.9% (47–68)	4.00	29	69.0% (51–83)	3.88
2023	113	73	53.4% (42–64)	0.33	40	55.0% (40–69)	1.18
2024	106	69	59.4% (48–70)	2.59	37	43.2% (29–59)	−1.90
2025	117	91	53.8% (44–64)	1.39	26	69.2% (50–83)	3.97
2026	92	68	47.1% (36–59)	0.96	24	70.8% (51–85)	4.29
All	644	469	55.9% (51–60)	2.41	175	61.7% (54–69)	2.24

Note: Signal days of the rule (a 15:50 body of 60% or more), 2021–2026. A day agrees when the sign of the summed dollar imbalance of the 50 largest stocks, from each stock's first closing-imbalance record received in the 15:50 minute, equals the candle's direction. Every signal day was classified, so no day is unclassified. Hit rate: share of days on which the CFD moved from the 15:51 open to the 15:59 close in the candle's direction, with its Wilson 95% interval in percent; bp: the mean of that move in the candle's direction, in basis points. Over all days the difference in hit rates is −5.9 points ($t = -1.35$), with a 20-day block-bootstrap 95% interval from −14.8 to +4.0 points.

more on agreeing days (+$80.43 against +$33.58 a trade), but not significantly ($t = 0.79$). By year (Table 42), disagreeing days had the higher hit rate in five of the six years, all but 2024.

In the words the plan fixed in advance, the prediction was not confirmed, the test has limited power, and a failure does not refute the auction explanation. The price may adjust within the 15:50 candle itself and leave little after 15:51, imbalances change after their first publication, the stocks' imbalances can offset one another in D, and 50 stocks are only part of the index. The high agreement is itself an association, not a cause: the candle and the imbalance can both reflect the same closing orders, such as index funds' orders known earlier in the day, and because records are timed by when Databento received them, the study cannot show that the publication came before the price move. What it does show is narrower: on these days the rule's edge was not concentrated on the days on which its candle agreed with the imbalance first published for the 50 largest stocks. Whether the effect changed when the NYSE moved its publication to 15:50 in 2019, and whether it depends on the size of the imbalance, were outside the plan.

16 Forward evidence

Table 43: Forward signals at real Robinhood prices, 22–30 October 2026.

Strike	trades	wins	total, hold to close	total, sell 15:59
ITM20	4	3	+$1,493.64	+$1,329.28
ITM15	4	3	+$1,193.64	+$1,084.28
ITM10	4	3	+$1,213.64	+$1,164.28
ITM5	4	3	+$658.64	+$484.28
ATM	4	2	−$69.36	−$173.72
OTM5	4	1	−$682.36	−$763.72
OTM10	4	0	−$529.36	−$565.72
OTM15	4	0	−$305.36	−$341.72

Note: Strong-body signal, four trades, $0.09 fee, dollars per contract. Four trades support no conclusion.

From 21 October to 6 November 2026, the strong-body candle fired on seven of thirteen sessions and was right on six. Only four of those days have real one-minute option prices; on the others the broker's minute bars had expired before they were collected. Table 43 prices the four trades at every strike. Three of the four signals were right, by 5 to 8 points, which was enough for in-the-money options but not for out-of-the-money ones, and selling at 15:59 was worse than holding at every strike. Four trades are too few to conclude anything, and at the recent effect size, confirming the option P&L would take many years of forward signals.

17 Discussion

17.1 What the evidence supports

After all of these studies the trading rule itself is unchanged, although the evidence behind it has changed. The rule, as tested, is to check at 15:51 whether the 15:50 one-minute candle has a body of at least 60% of its range and, if it does, to buy one SPXW 0DTE option about 10 points in the money in the candle's direction (a call 10 points below the rounded index after an up candle, a put 10 points above it after a down candle) and hold it to settlement.

The candle was always read from the CFD feed. The signal passed a pre-registered holdout on index moves, and at real asks the rule made money on 2021–2026, about two-thirds as much as the model implied. Two searches over 2,178 nearby settings found nothing reliably better, and a third over 7,128 settings that included other closing candles chose the 15:50 candle again. The re-validation found that the effect begins near a 60% body but cannot rank the thresholds between 60% and about 75%; beyond the threshold, a larger ratio was not followed by a larger gain. Deep in-the-money strikes were the most consistent, in the model and at real prices, and no early exit improved on holding.

The edge is small, about two index points over nine minutes, or roughly six cents per dollar of premium. It is concentrated in 2021–2022 and is not significant on its own since 2023. It is also fragile to execution: the bid-ask spread erased it before 2021, a one-minute delay costs a meaningful share of it, and on smaller contracts the spread exceeds it. Confirming the current rule does not show that it is the best rule, and two failed searches for something better do not show that nothing better exists.

The confirmation also applies only to the CFD candle. Read from the SPX index candle, which is what a trader sees, the rule made +$39.44 a trade against +$67.24 on the same days and did not meet its pre-registered criterion (Section 12). The index candle fires about as often as the tested rule only at a threshold of roughly 70%, and even above 70% the days on which it fired alone lost money. This is a practical problem as well as a statistical one, because in this project the CFD candle comes from daily history files that appear after the session, so the live alert could at first read only the index candle at 15:51.

The SPY candle partly solves that problem. Read from it, the rule made +$71.74 a trade, about what the CFD candle made on the same days, and met its retrospective criterion (Section 13); since that study the live alert reads the SPY candle too and acts on a rule built from it, the SPY candle at 70% or more with the index candle at 60% or more in the same direction. Because SPY was the third source tried on the same outcomes, and one of its lines was chosen after agreement between candles had been seen to do well, I treat this as a lead for the forward record and not as a confirmation. Across the three sources the pattern is the same. The gain came on sessions on which every candle showed a strong body, and a strong body on only some of them carried no edge.

The search of Section 14 is a comparison and not a test. No other closing minute, cut, direction or strike beat the 15:50 candle on 2021–2023, and the settings that did best over 2021–2026 were all 15:50 candles traded in their direction, most with a deeper strike than ITM10. Its own choice was not clearly beyond luck, and because it fell at 15:50 it could not be tested on the later years. Those settings are hypotheses for the forward record.

The test of Section 15 is about why the signal exists, not whether it pays. The candle usually points the same way as the closing-auction imbalance published at 15:50, but it was not more accurate on the days on which it did, so the most direct prediction of the auction explanation was not confirmed. The test could detect only a large difference, so that explanation is still open.

17.2 Negative results

This subsection lists the failures in one place. Of the 41 candle features, 38 did not qualify, and one of the three that did, a whole-day feature, failed out of sample. The plain 15:50 direction, without a body filter, predicts the close but not accurately enough to pay for an

option. The pre-registered skew rule failed for far out-of-the-money strikes. No early exit, momentum filter or dealer-gamma condition improved the trade, and all 26 real-time exit rules tested at real bids earned less than holding. At real asks the trade lost money before 2021. The first two grid searches failed to find a better setting, the second because its winner failed the luck diagnostic, and the third chose a 15:50 setting whose luck diagnostic missed its bar. A limit order at the mid did not capture the spread, and the trade was not confirmed on XSP or SPY. Read from the index candle, the rule did not meet its criterion, and its 60–70% band showed no edge, as on the SPY candle. The candle was not more accurate on days on which it agreed with the published closing-auction imbalance. Finally, the forward test lost three signal days because the broker's minute bars expired before they were collected.

17.3 Limitations

The design has several limitations, and the first is that there was no blind test. The passing candle overlaps a known effect, and the 2021–2026 index outcomes were reused by the real-price confirmation and by the later descriptive analyses.

The signal also comes from a CFD. Every trading decision uses a Dukascopy CFD candle and not the SPX index itself, and only the option prices and the settlement are real. I measured how much this matters. Read from the index candle, the rule made +$39.44 a trade and did not meet its criterion, the 60–70% band showed no edge, and the days on which only the index candle fired lost money (Section 12). That check reused outcomes that had already been seen. Read from the SPY candle, the rule met a retrospective criterion, but SPY was the third source tried on those outcomes, and a second vendor's SPY candle differed by a cent or two on about a third of the sessions compared (Section 13).

Execution adds further uncertainty. The real-price studies assume a fill at the quoted ask at the open of the minute, which ignores order size and fast markets. Trade bars show that the contract traded near the ask, but not the size or the queue, and the limit-order results are proxies that assume no delay. Entry is assumed at 15:51:00, while a person reacting to the candle trades seconds later. The trade also carries risk: a deep in-the-money contract moves almost point for point with the index, and at real asks the worst single trade on 2021–2026 lost $2,210 at ITM10.

The statistical tools and the samples have limits of their own. The luck diagnostic is approximate. Expiry frequency, volatility and pricing changed over the samples, so a circular shift is only an approximate benchmark for "no information", yet it decided the re-tuning study and was one of the gates of the final search. The early sample is thin: before 2016 SPXW had a same-day expiry only on some days, and daily expiries began in 2022, so tuning on 2013–2020 rests on a thin, uneven sample, and the 2021–2023 re-tuning on only three years. The strike and exit studies use a model calibrated on seven days of 2026 data, and the exit rules tested at real bids (Section 7.4) use one-minute snapshots and assume one contract and a firm bid one minute after the decision. Finally, the forward sample is small, with four priced forward trades.

18 Conclusion

I asked whether a one-minute candle near the close predicts the S&P 500 into the close, and whether the prediction can be traded with same-day options once the spread is paid.

One candle passed a pre-registered, out-of-sample test, the 15:50 one-minute SPX candle with a strong body, traded in its own direction to the close. It is a stricter version of an effect that was already known, and it has faded since 2023. Traded with a deep in-the-money 0DTE option bought at the quoted ask and held to settlement, it made money on 2021–2026, though less than a pricing model implied. The largest cost is the bid-ask spread, which took the whole edge before 2021, and a limit order at the mid does not recover it because it misses the best days. Re-tuned settings, early exits, momentum filters and smaller contracts did not improve on it, and a search over other closing candles chose the 15:50 candle again.

The evidence has limits. The edge is small, most of it comes from 2021–2022, and it is fragile to execution. The confirmation holds for the candle as the backtests read it, from a CFD that tracks the index. Read from the SPX index candle itself, under a plan locked in advance, the rule earned less and did not meet its criterion; the gain was confined to days on which both candles showed a strong body. Read from the SPY candle, which a trader can watch live, it met a retrospective criterion, with the gain again confined to sessions on which all three candles showed a strong body. A pre-registered test with the exchanges' imbalance messages found that the candle usually agrees with the closing-auction imbalance published at 15:50 but is no more accurate when it does, so that explanation of the signal was not confirmed.

One advantage of pre-registering every study is that the failed tests can be reported and used. Because each plan was fixed and hashed before the data it governs was read, a reader can see which refinements were tried and did not help, and which explanation the data did not confirm. To summarise, the signal passed a test at real prices when read from the CFD candle, did not pass when read from the index candle, and met a retrospective criterion when read from the SPY candle. It remains sensitive to execution. I think it is worth continuing to measure, but it is not an established edge.

There are three things I would do next. Settings suggested by the tuning results, such as the 15:48 candle, and the stricter 15:50 settings favoured by the final search should be treated as hypotheses and tested only on days after November 2026, under new registered plans. The forward record begun in November 2026 (Section 11), which since the SPY study holds all three candles and tracks the rule the alert now acts on, should run until it can test, on new days, whether sessions on which all three candles show a strong body continue to beat sessions on which only some do. The SPY candle's retrospective pass is a reason to keep that record and does not replace it, and the 80% cut on the index candle and the futures-lead explanation of Section 12 are hypotheses for it. Finally, forward collection should continue, with the number of forward days needed for a test fixed in advance, ideally alongside records of actual order fills, because only real fills can show how the trade should be executed.

Data and code availability. The analysis code, the pre-registration plans, their hash ledger and the shorter journal version of this paper are public at `https://github.com/kv1514/spx-closing-candle-study`. The OPRA data are licensed from Databento and are not redistributed.

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Appendix A: Pricing model calibration

This appendix sets out how the Bachelier model of Section 3.4 was calibrated and reports the model variants used in Section 5.

A.1 Volatility and skew

For a day with one-minute realised volatility RV, the root mean square of the 09:31–15:50 log returns, the model sets

$$\sigma' = k_{D1} \cdot RV \cdot 3 \cdot L, \tag{4}$$

where $3 = \sqrt{9}$ covers the nine minutes from 15:51 to 15:59 and $L = 7{,}709.58$ is the common index level. A strike's price is $f_j \cdot C$ or $f_j \cdot P$, where f_j is the skew factor for offset j. The rules came first and the values from the data. Implied volatility was fitted by least squares to the 15:51 opens of the out-of-the-money calls and puts within 30 points, on the seven days with real one-minute option bars (2026-10-20 to 10-30). k_{RH} is the median ratio of implied to realised volatility on Robinhood's SPX bars, c is the median ratio of Robinhood to CFD realised volatility over 22 overlapping days, and $k_{D1} = k_{RH}\, c$. The skew factor f_j is the median ratio of market to model price across days and sides (Table A1).

A.2 Model variants

Variant (a) is this calibration. Variant (b) uses a single constant volatility, variant (c) the scan's own cost assumption, and variant (d) calibrates k on 10-minute bars. The sensitivities change the half-spread, scale k, or drop the skew factors (Table A2). "Credible" variants in the text are all but (b), which ignores how volatile each day was.

Table A1: Calibrated skew factors.

	ITM20	ITM15	ITM10	ITM5	ATM	OTM5	OTM10	OTM15	OTM20
f_o	1.005	1.020	1.005	1.007	0.969	1.071	2.091	2.256	54.000
ratios	10	13	13	14	14	14	13	13	10

Note: "Ratios" is the number of market-to-model price ratios behind each factor. $k_{RH} = 1.089$; $c = 0.978$ over 22 days; $k_{D1} = 1.065$; ten-minute $k = 1.319$ over 22 days. The variant (a) model move to the close has a median of 10.10 bps (10th–90th percentile 5.72–19.20). Variant (c) uses $\sigma = 10.78$ points.

Table A2: Mean modelled P&L per contract in every variant.

Variant	ITM20	ITM15	ITM10	ITM5	ATM	OTM5	OTM10
(a) calibrated	+162	+143	+166	+161	+155	+92	−34
(b) constant vol	+192	+192	+242	+262	+261	+201	+127
(c) realized-move-fair	+186	+181	+158	+122	+83	+64	+62
(d) ten-minute k	+136	+104	+110	+85	+72	+12	−149
(a) half-spread \$0.05	+167	+148	+171	+166	+160	+97	−29
(a) half-spread \$0.20	+152	+133	+156	+151	+145	+82	−44
(a) $k \times 0.75$	+180	+172	+214	+235	+239	+170	+64
(a) $k \times 1.25$	+135	+102	+107	+81	+68	+8	−155
(a) no skew	+172	+174	+171	+166	+143	+106	+72

Note: Level-normalised dollars, \$0.09 fee, 644 modelled trades. Strikes 10–20 points in the money stay between +\$102 and +\$242; at the money ranges from +\$68 to +\$261.

Appendix B: Additional tables

This appendix reports three tables that support Sections 8 and 9.

B.1 The body threshold profile

Table B1 gives the profile behind Figure 4 in Section 8.

Table B1: The body threshold profile.

	2021–2023 (tuning)				2024–2026 (after the decision)			
Body at least	trades	\$/day	\$/trade	won	trades	\$/day	\$/trade	won
0.40	407	+7.23	+11.09	53%	468	+34.63	+51.57	51%
0.45	376	+12.60	+20.92	53%	437	+33.24	+53.02	51%
0.50	341	+9.08	+16.61	52%	393	+35.99	+63.82	52%
0.55	303	+15.60	+32.12	53%	353	+30.04	+59.32	52%
0.60 (current)	**270**	**+29.76**	**+68.78**	**55%**	**315**	**+29.97**	**+66.32**	**53%**
0.65	227	+26.38	+72.62	55%	281	+26.27	+65.15	53%
0.70	197	+25.90	+82.16	57%	238	+32.16	+94.20	55%
0.75	173	+28.43	+102.72	58%	207	+24.56	+82.71	53%
0.80	133	+37.84	+177.82	62%	167	+13.26	+55.33	50%
0.85	89	+25.51	+179.17	63%	130	+8.35	+44.77	48%
0.90	56	+11.19	+124.89	62%	95	+16.36	+120.02	55%

Note: 15:50 candle, with the candle, ITM10, real asks, \$0.09 fee, dollars per contract. Descriptive. The 2024–2026 columns were examined only after the re-tuning decision was recorded. At 0.85 and 0.90 the setting could not be eligible (fewer than 100 tuning trades).

B.2 Momentum filters

Table B2 reports the four momentum filters of Part B of the study in Section 9.

Table B2: Momentum filters on the candle trade (momentum study, Part B).

	trades		mean P&L / trade		mean difference from R0 per day			
Rule	kept	skipped	kept	skipped	$/day ($t$)	2013–16	2017–20	index (bps)
2013–2020: discovery, the selection test								
R0 (all)	905	0	+133.21	—	—	—	—	—
R1	490	415	+198.58	+56.02	−12.65 (−1.39)	+1.27	−24.45	−0.320
R2	471	434	+202.18	+58.36	−13.78 (−1.44)	+1.39	−26.64	−0.327
R3	376	529	+252.84	+48.18	−13.87 (−1.37)	+1.05	−26.51	−0.378
R4	707	198	+149.78	+74.04	−7.98 (−1.06)	+5.86	−19.70	−0.200
2021–2026: descriptive, not a test								
R0 (all)	644	0	+166.08	—	—			—
R1	312	332	+217.22	+118.02	−27.10 (−2.09)			−0.383
R2	315	329	+184.64	+148.31	−33.74 (−2.63)			−0.491
R3	225	419	+210.59	+142.18	−41.20 (−2.80)			−0.579
R4	458	186	+190.44	+106.09	−13.65 (−1.37)			−0.213

Note: ITM10, model variant (a), $0.09 fee, level-normalised dollars per contract. R0 takes every trade; R1 takes it only when it agrees with r_{rod}, R2 with the first half-hour return, R3 with both; R4 skips it on strong disagreement with r_{rod}. 2013–2020 uses 1,838 paired days and 2021–2026 uses 1,446. No rule was eligible, so no confirmation test was run.

B.3 Dealer gamma

Table B3 reports the two dealer-gamma tests of Part C of the same study.

Table B3: Dealer gamma (momentum study, Part C).

Test	Sample	Coefficient	t	p	Verdict
C1, d	2013–2026, 3,238 days	+0.022	0.95	0.34	not supported
C2, g	2013–2026, 1,549 trades	+$169	2.18	0.029	not supported
C2 guard (index, bps)	same trades	+3.13	2.54	0.011	—
C2, g	2013–2020, 905 trades	+$240	1.83		descriptive
C2, g	2021–2026, 644 trades	+$84	0.96		descriptive
Real asks, difference	2021–2026, 585 trades	−$16	−0.25[a]		descriptive

Note: C1: r_{last} on r_{rod}, the negative-gamma indicator N and $N \cdot r_{\text{rod}}$. C2: the candle trade's model P&L (ITM10, $0.09 fee) on N. Newey–West t; the proxy was negative on 30% of days. The real-ask row is descriptive and not in the plan; [a]two-sample t, trades independent.